

Nonabelian Support Obstructions

A Matrix Cousin Criterion
and the Surcomplex
Riemann–Hilbert Problem

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Abstract

We study one coefficient category from two sides. Its objects are matrices congruent to the identity modulo positive Hahn exponents, $I_r + \text{Mat}_r(\mathscr{J}_\Gamma)$, over a set-sized and possibly non-Archimedean ordered value group Γ , with ordinary holomorphic coefficient functions on a single fixed ordinary complex domain; when $\Gamma \subseteq \mathbf{No}$, the substitution $t^\gamma = \omega^{-\gamma}$ gives the surcomplex reading.

Part I solves the multiplicative gluing problem. On a puncture-and-disk cover of a connected open Riemann surface we prove an exact criterion: factor each transition matrix *independently* into a normalized polar factor and a factor holomorphic on the full disk, and test whether the union of the supports of the *polar* factors is well ordered. The criterion is invariant under arbitrary local holomorphic right gauges and under ordered value-group extension, and it produces genuinely nonabelian counterexamples: a rank-three family whose determinant is one, whose only raw singular exponent is 1, and whose abelianized problem splits, yet whose integral vector bundle is nontrivial; a repair of that family by a term which *worsens* its raw singular support; and, for every $r \geq 3$, an example trivial modulo the last central subgroup of the upper-unitriangular group but nonsplit in full positive rank r . We also prove support-controlled Stein splitting and deformation rigidity, with a cyclic-versus-noncyclic dichotomy.

Part II solves the inverse monodromy problem in the same category. A near-identity representation of $\pi_1(\mathbb{C} \setminus D)$ is the monodromy of a positive globally common-domain Hahn connection exactly when the union over punctures of the supports of $\rho(\ell_d) - I_r$ is well ordered; the realizer may then be taken coefficientwise logarithmic, and a fixed linear Mittag–Leffler period section selects a unique normalized realizer. We prove framed gauge classification, entire extension of logarithmic comparison gauges, a positive Frobenius form, a determinant-one refinement, calibrating examples on both sides of the criterion, an explicit mixed commutator correction in the value group $\mathbb{Q} + \mathbb{Q}\omega$, and the impossibility of a continuous linear period right inverse for infinitely many punctures.

The two headline criteria are different criteria on different objects. Part I tests the supports of normalized polar factors, not the raw transition supports; Part II tests the raw meridian monodromy supports. [Section 12](#) states which object each is a criterion on, and [section 8](#) contains the example that makes the distinction unavoidable.

Status and scope. This report merges two independently written manuscripts; see [appendix C](#). The exact global criteria and the explicit obstruction families are proposed new results in the specified Hahn-coherent categories. The local factorization mechanisms, the Neumann support calculus and every ordinary analytic input are established mathematics and are credited as such. The literature checks were targeted and certify no priority; no named published conjecture is claimed solved. Nothing here is refereed or machine checked. Nontriviality in Part I is proved over the *integral* coefficient sheaf only, and Part II assumes positive support and near-identity monodromy throughout: neither theorem is restated over a coefficient field in which negative-valuation changes of frame are admitted, and neither is the unrestricted classical Cousin or Riemann–Hilbert theorem. All infinite algebra is by well-ordered supports and finite coefficient multiplicity, never by convergence in the fine surreal topology.

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1 Two questions, one category

1.1 The shared setting

Let Γ be a set-sized totally ordered abelian group and let t^Γ be its monomial group. A coefficient of our objects is an ordinary holomorphic function on a *single fixed* ordinary complex domain, and an object is a Hahn series in t with well-ordered support whose coefficients all live on that same domain. The matrices we glue and the matrices we realize as monodromy both lie in

$$I_r + \text{Mat}_r(\text{positive-support part}),$$

so they are congruent to the identity modulo positive exponents. When Γ is an ordered additive subgroup of the surreal numbers $\mathbf{No}$, the Conway normal form substitution $t^\gamma = \omega^{-\gamma}$ interprets everything inside $\mathbf{SC} = \mathbf{No}[i]$. That interpretation is a reading of a Hahn model; it is never an assertion that a fine-topological surcomplex manifold or a fine-continuous surcomplex path is involved.

Two problems are asked in this one category.

Part I: multiplicative gluing (a matrix Cousin problem). Let X be a connected open Riemann surface, let $A \subset X$ be closed and discrete, let (D_a, u_a) be pairwise disjoint coordinate disks, and put $U_0 = X \setminus A$, $W_a = D_a \setminus \{a\}$. Given, *separately for each a* , a positive matrix G_a on W_a , when are there positive matrices B_0 on U_0 and B_a on D_a with

$$G_a = B_0 B_a^{-1} \quad \text{on } W_a? \tag{1.1}$$

Each local equation is harmless. The difficulty is that the entries of the *one global matrix* B_0 must have a legitimate, that is well ordered, Hahn support.

Part II: inverse monodromy (a Riemann–Hilbert problem). Let $D \subset \mathbb{C}$ be closed and discrete, $U = \mathbb{C} \setminus D$, and let $\rho : \pi_1(U, p) \rightarrow I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ be a near-identity representation. When is ρ the monodromy of a connection $dY = \Omega Y$ whose coefficients are ordinary holomorphic matrix one-forms on all of U and whose Hahn support is a single well-ordered positive set?

These are the same question asked multiplicatively and additively. Both need the local positive factorization at a puncture, both need an ordinary analytic input of Mittag–Leffler type applied one exponent at a time, both build their obstruction from a mixed term that the abelianization cannot see, and in both the answer is that a certain *union of Hahn supports must be well ordered*. That last coincidence of wording is also the one real trap in reading this report, so we state it immediately.

1.2 The two criteria, and why they are two

Criterion P (Part I, [theorem 6.2](#)). Gluing (1.1) is solvable if and only if

$$\mathcal{E}(G) = \bigcup_{a \in A} \text{supp}(\text{Pol}(G_a) - I_r) \quad \text{is well ordered,}$$

where $\text{Pol}(G_a)$ is the normalized polar factor produced by an *independent local factorization* of each transition matrix $G_a = \text{Pol}(G_a) \text{Reg}(G_a)$. This is a criterion on the polar factors, not on the transition matrices. It is *false* with $\bigcup_a \text{supp}(G_a - I_r)$ in place of $\mathcal{E}(G)$, in both directions.

Criterion M (Part II, [theorem 14.2](#)). A near-identity representation ρ is realizable if and only if

$$E_\rho = \bigcup_{d \in D} \text{supp}(\rho(\ell_d) - I_r) \quad \text{is well ordered,}$$

the union being over the matrices themselves, for a compatible system of based meridians ℓ_d freely generating $\pi_1(U, p)$. No factorization is applied first, and none may be: this is a criterion on the monodromy matrices, and it is basis- and basepoint-independent ([proposition 14.4](#)).

A reader who carries one “well-ordered support criterion” away from this report will test the wrong object in one of the two problems.

The distinction is not pedantic. [Section 8](#) contains a rank-three family over $\Gamma = \mathbb{Q}$ whose raw singular support is the single exponent $\{1\}$ and which nevertheless cannot be glued, because the normalized polar factors are forced to carry the descending set $\{1 + 1/n\}$; and it contains the opposite example, in which *adding* a term that introduces exactly that descending set among the raw poles *removes* the obstruction. So in Part I the raw singular support is neither sufficient nor necessary, and the factorization must be performed first. In Part II, by contrast, the raw monodromy support is exactly the invariant, and applying a factorization to $\rho(\ell_d)$ before testing would be a mistake of the opposite kind.

1.3 What each part contributes

Part I owns the machinery the two parts share: the integral common-domain Hahn sheaf on an ordinary base, the noncommutative support calculus, the normalized local polar factorization with right-gauge invariance, invariance under ordered value-group extension, the recovery of the known rank-one scalar test, and the Stein branch. Its own new content is the exact criterion $\mathcal{E}(G)$, the rank-three cross-term obstruction with its repair and its logarithm diagnostic, the arbitrary-depth central hierarchy, the resulting determinant-one integral vector bundles that are trivial on every bounded plane domain, the explicit localization boundary, and the cyclic-versus-noncyclic rigidity dichotomy.

Part II reads as the connection-and-monodromy application of that machinery, with its own new content: the three-way equivalence of [theorem 14.2](#) including *necessity*, which comes from a uniform support bound on holonomy valid for every group element; the fixed complex-linear period section that selects a unique normalized realizer; the framed gauge classification and the entire extension of logarithmic comparison gauges; the positive Frobenius form in which no spectral resonance operator is ever inverted; the SL_r refinement; three calibrating examples that pin the criterion from both

sides; the explicit first nonabelian correction with its $\mathbb{Q} + \mathbb{Q}\omega$ specialization; and the proof that an unrestricted infinite-puncture linear period right inverse cannot also be continuous.

1.4 Relation to reports already in the collection

This report is a continuation of the repository `VladimirReshetnikov/Surreal`, reviewed at commit

39f2be6667ade51bca2b45daa47e289d69c09764,

inspected September 21, 2026. Two companion reports are leaned on and are *cited rather than reproved*.

Global Support Obstructions: Moving Divisors, Mittag–Leffler, and a Picard Dichotomy, at `docs/surcomplex/global-divisors/article.tex` [2], proves the scalar support criteria and the Picard-group dichotomy for this workspace, and its subsection “Further precise directions” asks for exactly the present work: “the finite-rank vector-bundle theory: matrix-valued positive transition functions no longer reduce to an abelian logarithm identity, so a noncommutative support recursion must replace the rank-one argument”, and the same package over general ordinary open Riemann surfaces. Part I answers both. Its rank-one specialization *recovers* that report’s exact Cousin criterion, its Theorem 13.1, stated there in terms of the singular support $\Sigma(b)$; we cite that theorem as the scalar case rather than reprove it ([proposition 7.4](#)). Likewise the Picard-group vanishing dichotomy of that report — vanishing exactly when $\Gamma = \{0\}$ or $\Gamma = \mathbb{Z}\delta$ — is the rank-one shadow of our Stein dichotomy ([theorem 11.4](#)), driven by the same universal well-ordered set $\{\delta, 2\delta, \dots\}$.

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers, at `docs/surcomplex/differential-equations/article.tex` [3], contains the *forward* map that Part II inverts: its theorem “Monodromy valued in $\mathrm{GL}_d(K_\Gamma)$ ” constructs M_ℓ in $\mathrm{GL}_d(K_\Gamma)$ from a coherent connection, records that the coefficient of M_ℓ at exponent 0 is the ordinary monodromy matrix and that all other exponents lie in $S^* \setminus \{0\}$, and then states the categorical boundary we adopt verbatim rather than re-argue: “No fine-continuous surcomplex path is asserted: the loop is an ordinary complex path, continuation is coefficientwise, and the result is a matrix of Hahn constants.” [Theorem 14.2](#) is the converse of that map for near-identity representations, together with the exact condition on which ones arise.

1.5 What is and is not claimed as new

The Hahn support calculus is classical [7, 8]. The analytic interpretation of restricted ordinary functions over surreal workspaces is classical [9], and the broader landscape is surveyed in [10]. Algebraic Birkhoff factorization in complete filtered noncommutative algebras, including its recursive form and the necessity of commutator corrections, is established [12, 13], as are the ordinary Laurent, Mittag–Leffler, Behnke–Stein and Cartan inputs [14, 15, 16], iterated path integrals [18], and the classical regular-singular and Riemann–Hilbert background [19, 20]. We reprove only what we need with explicit arbitrary-ordered-group support control, and we claim no new general factorization principle and no new ordinary analytic theorem.

We do not classify bundles or connections beyond the stated questions. Part I decides triviality on a puncture-and-disk cover for arbitrary finite rank and proves a sufficient theorem for arbitrary Stein covers; Part II inverts holonomy for near-identity representations on a punctured plane. The boundaries in [section 21](#) are part of the theorem statements, not qualifications added afterwards.

2 The coefficient category

2.1 One set-sized workspace

For a complex algebra R , the Hahn algebra $R((t^\Gamma))$ consists of formal sums $F = \sum_{\gamma \in S} f_\gamma t^\gamma$ with $f_\gamma \in R$ whose nonzero support S is well ordered in the given order; addition is coefficientwise and multiplication is convolution. For matrices the support is the union of the supports of the finitely many entries, and an exponent occurs exactly when the corresponding coefficient matrix is not identically zero. The Hahn field of constants is

$$K_\Gamma = \mathbb{C}((t^\Gamma)), \quad \mathfrak{m}_\Gamma = \{a \in K_\Gamma : v(a) > 0\} \cup \{0\},$$

where v of a nonzero series is its least exponent. Divisibility of Γ and algebraic closedness of K_Γ are never used: we extract no roots and take no algebraic closure.

When $\Gamma \subseteq \mathbf{No}$ is an ordered additive subgroup, Conway normal forms give the interpretation

$$\mathbb{C}((t^\Gamma)) \longrightarrow \mathbf{No}[i] = \mathbf{SC}, \quad \sum c_\gamma t^\gamma \mapsto \sum c_\gamma \omega^{-\gamma}, \quad (2.1)$$

with real and imaginary parts interpreted separately [9, 8, 11]. All supports here are sets; no sum indexed by a proper class, no proper-class fundamental group and no proper-class coefficient ring occurs anywhere.

All ordinary manifolds and Riemann surfaces carry their usual Hausdorff, second-countable topology. For a connected ordinary complex domain V set

$$\mathcal{H}_\Gamma(V) = \mathcal{O}(V)((t^\Gamma)), \quad (2.2)$$

$$\mathcal{I}_\Gamma(V) = \{F \in \mathcal{H}_\Gamma(V) : \text{supp } F \subseteq \Gamma_{\geq 0}\}, \quad (2.3)$$

$$\mathcal{J}_\Gamma(V) = \{F \in \mathcal{H}_\Gamma(V) : \text{supp } F \subseteq \Gamma_{> 0}\}. \quad (2.4)$$

Every coefficient must be holomorphic on the *same fixed domain* V ; we never substitute a family of convergent germs with unrelated radii, and $\mathcal{H}_\Gamma(V)$ is in general *not* the algebraic tensor product $K_\Gamma \otimes_{\mathbb{C}} \mathcal{O}(V)$, since a legitimate family may involve infinitely many linearly independent coefficient functions. For disconnected open sets, sections are specified component by component. Writing $\Omega_{\text{hol}}^1(V)$ for ordinary holomorphic one-forms, the same notation with one-form coefficients denotes one-form families. There is a reduction map

$$\text{red} : \mathcal{I}_\Gamma \longrightarrow \mathcal{O}, \quad \sum_{\gamma \geq 0} f_\gamma t^\gamma \mapsto f_0, \quad \ker \text{red} = \mathcal{J}_\Gamma, \quad (2.5)$$

and the positive matrix group is

$$\mathcal{G}_{r,\Gamma}(V) = I_r + \text{Mat}_r(\mathcal{J}_\Gamma(V)). \quad (2.6)$$

Its elements are automatically two-sided invertible by [lemma 3.4](#). Throughout, “positive” refers to *Hahn exponents only*: never to positivity of a Hermitian matrix, never to positivity of entries, and never to positive-definiteness. Elements of $I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ need not be algebraically unipotent.

2.2 Two standing conventions that differ between the two parts

The two parts of this report were developed independently and do not make identical structural assumptions. Both assumptions are correct as stated and neither is weakened here.

Part I *proves and uses* the sheaf property of these assignments on the ordinary base. The point is a support point. A nonzero ordinary holomorphic coefficient on a connected domain cannot vanish on a nonempty open subset, so restriction between nonempty connected ordinary domains preserves support. If compatible local sections are given on a cover of a connected domain, the identity theorem along chains of intersecting patches shows that their coefficient functions glue and that the global support equals the support seen on any one nonempty connected patch; that support is well ordered. For disconnected opens one takes the product over connected components. In particular, if a section on $X \setminus A$ extends separately across each point of a discrete A , it extends *with its existing global support*: no new exponent can appear only at an isolated ordinary point. This observation is used whenever a matrix quotient is extended across a puncture.

Part II deliberately does *not* assume that the global-support assignment is a sheaf for arbitrary covers, and none of its proofs uses that. All its gluing is done directly, by descending families already supported in one specified monoid, and its only analytic input is the ordinary Mittag–Leffler theorem applied one exponent at a time. Consequently no application of a Cartan-type theorem to an infinite-dimensional Hahn coefficient sheaf is hidden in Part II, and no companion claim about Hahn Noetherianity or a Hahn Nullstellensatz is relied upon anywhere in this report.

2.3 Comparison of names across the two parts

The following table fixes one vocabulary. Where the two source manuscripts used different names for the same object, the merged name is given first; where they used the same word for genuinely different objects, the objects are separated.

Merged name	Part I usage	Part II usage
$\mathcal{H}_\Gamma(V), \mathcal{I}_\Gamma(V), \mathcal{J}_\Gamma(V)$	full, integral and positive common-domain sheaves; sheaf property proved on the ordinary base	$\mathcal{J}_\Gamma(V)$ is the positive part of the common-domain family; no sheaf property assumed
$\mathcal{G}_{r,\Gamma}(V) = I_r + \text{Mat}_r(\mathcal{J}_\Gamma(V))$	the positive matrix group being split	the group of positive framed gauges, and the target of ρ after restriction to one domain
$\mathfrak{m}_\Gamma$	not used; positivity is imposed sheaf-wise	positive-valuation elements of K_Γ ; ρ takes values in $I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$
Well-ordered union of supports	taken over normalized <i>polar factors</i> $\text{Pol}(G_a)$: the invariant $\mathcal{E}(G)$	taken over the <i>monodromy matrices</i> $\rho(\ell_d)$: the invariant E_ρ
Local positive factorization	$G = \text{Pol}(G) \text{Reg}(G)$ on a punctured disk, normalized Birkhoff type	$Y = H \exp(R \text{Log } z)$ on a punctured disk, positive Frobenius type
$\Sigma(\cdot)$	the rank-one logarithmic singular support, used only for the scalar comparison	not used
Strong summability	implicit in Hahn summability of finite expressions	a named condition on set-indexed families; distinguished sharply from valuation-topology convergence

Remark 2.1 (Resonance does not occur in this report). Neither part imposes, or needs, any arithmetic or small-divisor hypothesis, and the word *resonance* appears here only as an absence: in [theorem 5.4](#) no spectral resonance operator is inverted, because positivity forces the exponent-zero part of the residue to vanish. Other reports in this collection use “resonance” in several mutually inequivalent senses; none of them is in force here, and no theorem of this report may be combined with one of those hypotheses by analogy.

Remark 2.2 (Which exponential and which logarithm). Every exp and log in this report is the formal one-variable substitution of [lemma 3.4](#), applied to a matrix with positive Hahn support and computed coefficientwise on a fixed ordinary domain. Its domain is $\text{Mat}_r(\mathcal{J}_\Gamma(V))$, and the only law used is that exp and log are mutually inverse as one-variable substitutions. In particular $\log(AB) \neq \log A + \log B$ for noncommuting arguments — this failure is the entire difficulty in Part I — and no exponentiation here is the Ehrlich–Kaplan surcomplex exponential, or any radial or quaternionic extension of it, whose homomorphism properties are separate questions belonging to separate reports. The scalar identity $\text{Pol}(G) = \exp(\pi_- \log G)$ of [\(5.5\)](#) is used only in rank one, and [proposition 8.4](#) shows what happens if one tries to use it in rank three.

2.4 Surcomplex evaluation on a halo

At an ordinary point $z_0 \in V$ and an $\varepsilon \in \mathfrak{m}_\Gamma$, a positive family is evaluated by the ordinary Taylor series of its coefficients:

$$F^\sharp(z_0 + \varepsilon) = \sum_{\gamma} \sum_{k \geq 0} t^\gamma \frac{F_\gamma^{(k)}(z_0)}{k!} \varepsilon^k, \quad (2.7)$$

the exponent-zero term being allowed separately in the integral case. If S is the Hahn support and $T = \text{supp } \varepsilon$, the resulting support lies in $S + \langle T \rangle_{\mathbb{N}}$, and each fixed exponent receives only finitely many contributions by [lemma 3.2](#). So our matrices do give matrices on finite surcomplex thickenings of ordinary charts, entry by entry. This is offered only to explain why the category deserves to be called surcomplex. The ordinary complex base is not being replaced by a fine-topological surcomplex manifold, and no monodromy or gluing statement below is obtained by transport along a fine-continuous surcomplex loop.

2.5 Ordinary paths, not fine-surreal paths

For a piecewise smooth ordinary path c in an ordinary domain, integrals of coefficient one-forms and their iterated integrals are ordinary complex integrals; their values are then assembled as Hahn coefficients. Paths, the universal cover, iterated integrals and Mittag–Leffler are all ordinary complex. Only the assembly of coefficients is Hahn. We assert no convergence of partial Hahn sums in the fine topology of $\mathbf{SC}$, and [example 17.3](#) shows that such partial sums can fail to converge even in the valuation topology. This is the same domain separation stated in the repository’s differential-equations report [\[3\]](#), whose wording we adopt rather than re-argue: the loop is an ordinary complex path, continuation is coefficientwise, and the result is a matrix of Hahn constants.

3 Support calculus without any convergence assumption

Definition 3.1 (Strong summability). A set-indexed family of Hahn series, or of matrices of Hahn series, is *strongly summable* if the union of its supports is well ordered and each exponent occurs in the support of only finitely many members. Its sum is then the finite coefficientwise sum at each exponent. This is not valuation-topology convergence, which is never used below.

Lemma 3.2 (Neumann support calculus; classical input). *Let $E \subseteq \Gamma_{>0}$ be well ordered and let $M = \langle E \rangle_{\mathbb{N}} = \{0\} \cup \{e_1 + \cdots + e_n : n \geq 1, e_j \in E\}$. Then M is well ordered, and for each $\gamma \in \Gamma$ only finitely many finite words in E , across all word lengths, have sum γ . Consequently, if $S \subseteq \Gamma$ is well ordered then $S + M$ is well ordered, a fixed element has only finitely many decompositions with*

one summand in each of two well-ordered sets, and every fixed coefficient of an expression formed by adjoining finitely many positive-support factors receives only finitely many contributions.

This is the positive-support form of Neumann’s lemma [7], with the maximally complete field background in [8]. We use it as an established combinatorial input and do not rename it. It is strictly stronger than the statement that all exponents are positive, and the finite-word clause across all word lengths is exactly what powers noncommutative Dyson expansions and geometric inverses below. Both source manuscripts of this report state and use it independently, in the same form.

Remark 3.3 (No hidden Archimedean or cofinality assumption). A least positive input exponent δ need not have the property that $n\delta$ eventually exceeds a prescribed γ . In $\mathbb{Z}^2$ with lexicographic order, $n(0, 1) < (1, 0)$ for every ordinary integer n . In $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ with its induced surreal order, $n < \omega$ for every ordinary positive integer n , so repeatedly increasing a valuation by 1 never reaches ω ; nevertheless $\sum_{n \geq 0} t^n$ is a valid Hahn sum and the support calculus works equally well at the exponents ω and $\omega + 1$, which is where the correction of [proposition 18.1](#) will sit. No proof below infers coefficient finiteness from that false cofinality. Every recursion is a well-founded recursion on a well-ordered support set.

Lemma 3.4 (Positive inversion and formal functions). *Let V be an ordinary domain and $A \in \text{Mat}_r(\mathcal{J}_\Gamma(V))$, with $\text{supp } A \subseteq E$ for a positive well-ordered E . Then*

$$(I_r + A)^{-1} = \sum_{k \geq 0} (-A)^k, \quad \exp A = \sum_{k \geq 0} \frac{A^k}{k!}, \quad \log(I_r + A) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} A^k$$

are strongly summable coefficientwise on V , with nonconstant supports in $\langle E \rangle_{\mathbb{N}} \setminus \{0\}$. The inverse is two-sided, and $\exp$ and $\log$ are mutually inverse as one-variable formal substitutions. They do not make matrix multiplication additive. The same statements hold with coefficients in any associative complex algebra for which the finitely many operations in each displayed coefficient are defined.

Proof. Expand an entry of A^k : its terms are indexed by a word of k positive exponents together with a path through r matrix indices. At a fixed total exponent only finitely many exponent words occur by [lemma 3.2](#), and each word admits only finitely many matrix-index paths. Hence every displayed coefficient sum is finite and its value is holomorphic on the original V , and all supports lie in $\langle E \rangle_{\mathbb{N}}$. Ordinary formal identities in one variable, and the geometric-series cancellation, may therefore be checked coefficientwise, because at a fixed exponent all sufficiently long words fail to contribute. Noncommuting independent arguments do not satisfy the scalar exponential addition law. $\square$

Lemma 3.5 (Finite decomposition inside the generated monoid). *Put $S = \langle E \rangle_{\mathbb{N}} \setminus \{0\}$ for a positive well-ordered E . Each $\gamma \in \Gamma$ has only finitely many ordered decompositions into members of S , and only finitely many members of S can occur as a part in such a decomposition of a fixed γ .*

Proof. Choose an E -word representing each member of S . Flattening a word in S of total weight γ gives an E -word of that weight, and there are finitely many such by [lemma 3.2](#). A fixed finite word has only finitely many partitions into nonempty consecutive blocks, and every original S -word is recovered from one of these finitely many possibilities. Taking all parts of those words gives the second assertion. $\square$

[Lemma 3.5](#) is what makes the effectivity statement [proposition 19.1](#) possible; it is not needed for [lemma 3.4](#).

4 The ordinary analytic inputs

Everything analytic in this report is an ordinary complex theorem applied at *one Hahn exponent at a time*. That is the reason the criteria exist at all: the ordinary problems are always solvable, and the obstruction is that their outputs across exponents must assemble into one legitimate Hahn section. The two parts use two different packagings of the ordinary input, and both are kept, because each theorem below uses its own.

4.1 A fixed-disk Laurent splitting

Fix $D = \{|u| < \rho\}$ and $D^* = D \setminus \{0\}$. A function $f \in \mathcal{O}(D^*)$ has the Laurent decomposition

$$f = f_- + f_+, \quad f_- = \sum_{k \geq 1} a_{-k} u^{-k}, \quad f_+ = \sum_{k \geq 0} a_k u^k. \quad (4.1)$$

Write $\mathcal{P}$ for the algebra of all such negative parts and π_-, π_+ for the projections, acting entrywise on matrices.

Lemma 4.1 (The ordinary splitting algebras). *There is a direct-sum decomposition of complex vector spaces $\mathcal{O}(D^*) = \mathcal{P} \oplus \mathcal{O}(D)$. Both summands are closed under multiplication, although $\mathcal{P}$ has no multiplicative identity; their intersection is zero; and the decomposition does not shrink D .*

Proof. For every $s \in (0, \rho)$, Cauchy’s formula bounds the negative Laurent coefficients by $|a_{-k}| \leq M_s s^k$, so the series in $1/u$ converges for $|u| > s$ with s arbitrarily small, and vanishes at infinity. Conversely a function holomorphic on $\mathbb{C} \setminus \{0\}$ and vanishing at infinity has only negative Laurent powers. Products retain these properties. A function in the intersection extends across zero and across infinity, hence is constant on the sphere with value zero at infinity. The positive part converges on the original disk of radius ρ . $\square$

The negative part may have an essential singularity at zero; no bound on pole orders is imposed. The explicit obstruction families below use only simple poles.

4.2 Isolated singular parts: the Part I packaging

Lemma 4.2 (Isolated singular parts on the plane). *Let $A \subset \mathbb{C}$ be closed and discrete and let p_a be a prescribed negative Laurent part in $z - a$ at each $a \in A$. There exists $f \in \mathcal{O}(\mathbb{C} \setminus A)$ such that $f - p_a$ extends holomorphically through each disk D_a . One may additionally prescribe $f(z_*)$ at one fixed $z_* \notin A$.*

Proof. The finite case is a finite sum. Otherwise enumerate $A = \{a_n\}$ with $|a_n| \rightarrow \infty$ and choose radii $R_n \rightarrow \infty$ with $R_n < |a_n|$ for all large n . Each $p_{a_n}(z - a_n)$ is holomorphic on $|z| < |a_n|$; choose a Taylor polynomial q_n at zero with $\sup_{|z| \leq R_n} |p_{a_n}(z - a_n) - q_n(z)| < 2^{-n}$. After separating finitely many early indices, $f(z) = \sum_n (p_{a_n}(z - a_n) - q_n(z))$ converges normally on every compact subset of $\mathbb{C} \setminus A$. Near a fixed a_m all terms except the m th form a normally convergent holomorphic series, so the required negative part is p_{a_m} , and the differences extend over the full D_{a_m} by uniqueness. Adding a constant fixes $f(z_*)$. $\square$

The proof allows essential isolated singularities, because it approximates the entire negative part away from its centre, and uses no bound on pole orders. The same correcting-polynomial mechanism appears in the scalar Cousin step of the repository’s global-divisor report [2].

Lemma 4.3 (Open-surface version). *The conclusion of lemma 4.2 holds on any connected open Riemann surface X , with A closed and discrete, pairwise disjoint coordinate disks D_a , and negative Laurent parts in the chosen coordinates.*

Proof. An open Riemann surface is Stein by the Behnke–Stein theorem [14]. The data p_a define an ordinary additive holomorphic cocycle on the cover $X \setminus A$, D_a . Cartan’s theorem B gives $H^1(X, \mathcal{O}) = 0$ and hence a splitting of that cocycle; in the chosen sign convention the central component is the desired f . A trivial additive torsor has a section expressed in the original charts, so no acyclicity assumption on this particular cover is needed. A constant again prescribes one value. See [15, 16] for the ordinary coherent-sheaf input. $\square$

On the plane, lemma 4.2 makes the analytic existence step of theorem 6.2 self-contained. The general open-surface statement uses exactly the two established ordinary theorems named in its proof, and not an unproved theorem B for a Hahn coefficient sheaf.

4.3 The period map and a linear section: the Part II packaging

Let $D \subset \mathbb{C}$ be closed and discrete. Every compact set meets D in a finite set, so D is at most countable, and $U = \mathbb{C} \setminus D$ is connected.

Lemma 4.4 (Free based meridians). *There is a collection of based positively oriented meridians $(\ell_d)_{d \in D}$ forming a free basis of $\pi_1(U, p)$; every group element is a finite word in these meridians and their inverses.*

Proof. Exhaust $\mathbb{C}$ by nested disks centred at p with boundaries missing D . Each disk contains finitely many punctures and has free fundamental group generated by based meridians. When the disk is enlarged, the old meridians can be retained and meridians for the new punctures added: cutting the annular region along arcs to the newly included punctures, or applying van Kampen, exhibits the old group as a free factor. Choose these bases compatibly. A loop and any specified homotopy have compact image, hence occur in a sufficiently large disk, so the fundamental group of the union is free on the union of the compatible bases. For $D = \emptyset$ the group is trivial. $\square$

We fix such a basis throughout Part II. The numerical matrices assigned to based meridians depend on the chosen tails and on the base point, and that dependence is not suppressed. So that $dY = \Omega Y$ yields a homomorphism for matrix multiplication, $\alpha\beta$ means traversal of β first and then α . All meridians are positively oriented.

Let $\mathcal{L}(D)$ be the complex vector space of meromorphic one-forms on $\mathbb{C}$ with at most simple poles at the points of D and no other finite poles. There is *no condition at infinity*. Define

$$P : \mathcal{L}(D) \longrightarrow \mathbb{C}^D, \quad (P\alpha)_d = \int_{\ell_d} \alpha = 2\pi i \operatorname{Res}_d \alpha. \quad (4.2)$$

The target is the full *product* of complex vector spaces, not the direct sum. That is the whole reason no summability across punctures is needed in theorem 14.2.

Lemma 4.5 (A linear period right inverse). *The map P is surjective, and there is a complex-linear $\sigma : \mathbb{C}^D \rightarrow \mathcal{L}(D)$ with $P\sigma = \operatorname{id}$. For finite D one may take*

$$\sigma((v_d)_{d \in D}) = \frac{1}{2\pi i} \sum_{d \in D} v_d \frac{dz}{z - d}. \quad (4.3)$$

For $D = \emptyset$ the right inverse has zero image.

Proof. Prescribing the principal part $v_d dz/(2\pi i(z-d))$ at each d and applying the ordinary Mittag–Leffler theorem proves surjectivity; the elementary mechanism is the correcting-polynomial argument of [lemma 4.2](#), applied to the one-form data. That construction need not be linear in the input, because the chosen polynomial degrees may depend on it. To obtain a linear section, split the surjection of complex vector spaces: choose a basis of $\mathbb{C}^D$, choose one preimage of each basis element, and extend linearly. This uses the axiom of choice for vector spaces. For finite D the displayed finite sum is an explicit linear section. $\square$

Remark 4.6 (What is and is not required of the section). For infinite D we assert neither continuity nor a norm estimate nor an effective algorithm for σ ; [proposition 19.3](#) shows that continuity is in fact impossible for the natural topologies. In particular $\sum_d v_d/(z-d)$ cannot simply be written without a convergence correction. The theorem needs only an ordinary complex-linear splitting. The section is chosen once, independently of the Hahn exponent, and is applied *entrywise* to matrix data; that last point is what makes the traceless refinement [theorem 16.3](#) true.

Both packagings are the ordinary Mittag–Leffler theorem, and neither is derived from the other here. [Lemma 4.2](#) gives, for each exponent separately, one function with prescribed negative parts of unbounded order; [lemma 4.5](#) gives a single *linear* choice of simple-pole one-form, valid simultaneously for all exponents, at the cost of the axiom of choice and of continuity. Part I needs the first because its prescribed local data are arbitrary negative Laurent matrices; Part II needs the second because uniqueness of its normalized realizer is exactly injectivity of P on $\text{im } \sigma$.

5 Local positive factorizations at a puncture

Two local normal forms are used, one per part. They are different normalizations of different objects and neither is a consequence of the other.

5.1 The normalized polar factorization

For the chosen disk and coordinate put

$$\mathcal{P}_{r,\Gamma} = I_r + \text{Mat}_r(\mathcal{P}((t^\Gamma))_{>0}),$$

meaning that the nonconstant coefficients lie in $\mathcal{P}$ and have positive well ordered support. This is a group: nonconstant coefficients of a product or of a positive inverse are finite sums of products of negative Laurent parts, and each such product still vanishes at auxiliary infinity.

Theorem 5.1 (Local normalized factorization). *Let $G \in \mathcal{G}_{r,\Gamma}(D^*)$ and $S = \text{supp}(G - I_r)$. There exist unique matrices*

$$P \in \mathcal{P}_{r,\Gamma}, \quad H \in \mathcal{G}_{r,\Gamma}(D), \quad G = PH, \tag{5.1}$$

and both nonconstant supports lie in $M = \langle S \rangle_{\mathbb{N}} \setminus \{0\}$. We write $P = \text{Pol}(G)$ and $H = \text{Reg}(G)$. Every coefficient of H is holomorphic on the original disk, not on an exponent-dependent smaller disk.

Proof. The case $S = \emptyset$ is immediate. Otherwise write $G = I_r + \sum_{\gamma > 0} g_\gamma t^\gamma$, $P = I_r + \sum_{\gamma \in M} p_\gamma t^\gamma$, $H = I_r + \sum_{\gamma \in M} h_\gamma t^\gamma$, with $g_\gamma = 0$ at exponents not originally present, and use well-founded recursion

on M . Once all smaller coefficients are chosen, define

$$r_\gamma = g_\gamma - \sum_{\substack{\eta+\theta=\gamma \\ \eta, \theta > 0}} p_\eta h_\theta, \quad (5.2)$$

$$p_\gamma = \pi_-(r_\gamma), \quad h_\gamma = \pi_+(r_\gamma). \quad (5.3)$$

Every summand involves strictly smaller exponents and the sum is finite by [lemma 3.2](#), so each step produces a negative Laurent matrix and a matrix holomorphic on D . The series have supports in M and satisfy $PH = G$ coefficientwise; their inverses exist by [lemma 3.4](#).

For uniqueness, suppose $P_1 H_1 = P_2 H_2$, so $P_2^{-1} P_1 = H_2 H_1^{-1}$. The nonconstant coefficients on the left lie in $\mathcal{P}$ and those on the right are holomorphic on D . After placing the finitely many input supports in a common positive generated monoid, coefficientwise comparison and [lemma 4.1](#) force both sides to equal I_r . $\square$

Proposition 5.2 (Right-gauge invariance). *If $Q \in \mathcal{G}_{r,\Gamma}(D)$ then*

$$\text{Pol}(GQ) = \text{Pol}(G). \quad (5.4)$$

The analogous statement for multiplication on the left is generally false.

Proof. The factorization $GQ = \text{Pol}(G)(\text{Reg}(G)Q)$ has the defining properties of [theorem 5.1](#), and uniqueness gives the equality. Left multiplication can mix regular and polar terms in the opposite order; the examples of [section 8](#) show that the order matters. $\square$

Corollary 5.3 (Unitriangular factors). *If G is upper unitriangular then so are $\text{Pol}(G)$ and $\text{Reg}(G)$. If a fixed pattern of strictly upper-triangular entries is closed under matrix multiplication, the factors preserve that pattern too.*

Proof. Both Laurent projections preserve zero entries, and every nonlinear term in (5.2) is a product of previously constructed matrices in the specified multiplicatively closed pattern. Induction finishes the proof. $\square$

5.2 Where the obstruction is manufactured

For a single positive parameter, writing $G = I_r + tg_1 + t^2g_2 + t^3g_3 + \cdots$, the recursion begins

$$\begin{aligned} p_1 &= \pi_-g_1, & h_1 &= \pi_+g_1, \\ p_2 &= \pi_-(g_2 - p_1h_1), & h_2 &= \pi_+(g_2 - p_1h_1), \\ p_3 &= \pi_-(g_3 - p_1h_2 - p_2h_1). \end{aligned}$$

The second-order correction already contains the product of a polar part and a regular part, and it can be nonzero even when $g_2 = 0$. With different positive exponents at different punctures, the exponents of such products can form a descending sequence absent from the raw singular data. That single mechanism is the whole content of [theorem 8.2](#) and [theorem 9.2](#).

In rank one, and only there, the factorization has a closed form:

$$\text{Pol}(G) = \exp(\pi_- \log G), \quad \text{Reg}(G) = \exp(\pi_+ \log G), \quad (5.5)$$

with the projections acting coefficientwise and the scalar exponential addition law verifying the factorization. In a noncommutative matrix algebra, replacing [theorem 5.1](#) by (5.5) is invalid; the

algebraic Birkhoff literature describes the associated Baker–Campbell–Hausdorff corrections in a filtered setting [12, 13], and our direct recursion avoids expanding that formula. [Proposition 8.4](#) exhibits the exact numerical discrepancy in rank three.

5.3 The positive Frobenius form

The next theorem is the local normal form used in Part II. It explains why residues still control *local* monodromy while being unreadable directly from the prescribed *based* monodromy matrices. It is a support-controlled version of the elementary nonresonant regular-singular construction, not a new claim about classical Frobenius theory.

Theorem 5.4 (A common-disk positive Frobenius factorization). *Let Δ be an ordinary disk about 0 and suppose a positive Hahn connection on $\Delta \setminus \{0\}$ has the form*

$$\Omega = R \frac{dz}{z} + B(z) dz, \quad (5.6)$$

with $R \in \text{Mat}_r(\mathfrak{m}_\Gamma)$ and $B \in \text{Mat}_r(\mathcal{J}_\Gamma(\Delta))$ sharing one positive well ordered support E . Then there is a unique $H \in I_r + \text{Mat}_r(\mathcal{J}_\Gamma(\Delta))$ with $H(0) = I_r$ such that, on a branch of the logarithm,

$$Y(z) = H(z) \exp(R \text{Log } z) \quad (5.7)$$

solves $dY = \Omega Y$. The supports of $H - I_r$ and $H^{-1} - I_r$ lie in $\langle E \rangle_{\mathbb{N}} \setminus \{0\}$, and the local monodromy based at $z_0 \in \Delta \setminus \{0\}$ is

$$H(z_0) \exp(2\pi i R) H(z_0)^{-1}. \quad (5.8)$$

Proof. The required equation is $H' = [R, H]/z + BH$ with $H(0) = I_r$. Construct its coefficients recursively on $\langle E \rangle_{\mathbb{N}} \setminus \{0\}$: at weight γ one must solve

$$H'_\gamma = B_\gamma + \sum_{\substack{\alpha+\beta=\gamma \\ \alpha, \beta > 0}} \left(B_\alpha H_\beta + \frac{[R_\alpha, H_\beta]}{z} \right), \quad H_\gamma(0) = 0,$$

a finite sum. The previously constructed H_β vanish at 0, so H_β/z is holomorphic on the entire disk; the right-hand side is therefore holomorphic on that same disk, and integration from 0 defines a unique holomorphic H_γ vanishing there. The resulting family has the claimed support and its inverse is given by [lemma 3.4](#). A least-difference argument proves uniqueness.

Because R has positive support, $\exp(R \text{Log } z)$ is a strongly summable family of ordinary holomorphic functions on any logarithm branch, and coefficientwise differentiation gives $(\exp(R \text{Log } z))' = R \exp(R \text{Log } z)/z$; the equation for H then verifies (5.7). Continuing once positively around 0 increases $\text{Log } z$ by $2\pi i$ and multiplies the fundamental matrix on the right by $\exp(2\pi i R)$. Normalizing at z_0 conjugates that factor by $H(z_0) \exp(R \text{Log } z_0)$, and the exponential factor commutes with $\exp(2\pi i R)$ and cancels, giving (5.8). $\square$

There is no division by a spectral resonance operator in this proof: the exponent-zero coefficient of the residue is zero, so the equation at each new exponent is an ordinary primitive problem. Positivity is doing real work here (compare [remark 2.1](#)).

If a meridian is based at a distant point p , its tail contributes a further transport conjugation to (5.8), and with several punctures these conjugations interact. Choosing $R_d = (2\pi i)^{-1} \log M_d$

separately at each puncture therefore need not solve the based inverse problem. The explicit failure appears as a mixed commutator term in [proposition 18.1](#).

Part I. Multiplicative gluing: the polar-support criterion

6 The sharp global matrix Cousin criterion

Let X be a connected open Riemann surface, $A \subset X$ closed and discrete, and choose pairwise disjoint coordinate disks (D_a, u_a) with $u_a(a) = 0$. Put $U_0 = X \setminus A$ and $W_a = D_a \setminus \{a\}$; the punctured surface U_0 is connected. For each a , *independently*, let $G_a \in \mathcal{G}_{r,\Gamma}(W_a)$. The disks do not meet one another, so this is a complete positive matrix cocycle on the displayed cover, with the inverse transition on the reversed overlap. There need not be one well-ordered set containing the supports of all the G_a .

Definition 6.1 (Normalized polar support). Factor $G_a = P_a H_a$ by [theorem 5.1](#) and define

$$\mathcal{E}(G) = \bigcup_{a \in A} \text{supp}(P_a - I_r) \subseteq \Gamma_{>0}. \quad (6.1)$$

The datum is *polar-admissible* when $\mathcal{E}(G)$ is well ordered. No sum over a is intended in (6.1): it is a set-theoretic union of supports, taken across all matrix entries.

Theorem 6.2 (Exact nonabelian Cousin criterion: Criterion P). *The following are equivalent.*

- (i) *There exist $B_0 \in \mathcal{G}_{r,\Gamma}(U_0)$ and $B_a \in \mathcal{G}_{r,\Gamma}(D_a)$ with $G_a = B_0 B_a^{-1}$ on every W_a .*
- (ii) *The set $\mathcal{E}(G)$ is well ordered.*

When these hold, let $M = \langle \mathcal{E}(G) \rangle_{\mathbb{N}}$. One can choose

$$\text{supp}(B_0 - I_r) \subseteq M \setminus \{0\}, \quad (6.2)$$

and in the local factorizations $B_0 = P_a Q_a$ the matrices $Q_a - I_r$ can also be supported in $M \setminus \{0\}$. The disk trivializers are $B_a = H_a^{-1} Q_a$ and hence have individually legitimate supports. A value $B_0(z_*) = I_r$ may be imposed at any chosen $z_* \in U_0$.

Proof. Necessity. Suppose $G_a = B_0 B_a^{-1}$. By right-gauge invariance (5.4),

$$P_a = \text{Pol}(G_a) = \text{Pol}(B_0|_{W_a}).$$

Let $S = \text{supp}(B_0 - I_r)$. The local support estimate in [theorem 5.1](#) is independent of the puncture and of the Laurent coefficients of the input, so $\text{supp}(P_a - I_r) \subseteq \langle S \rangle_{\mathbb{N}} \setminus \{0\}$ for every a . That one generated monoid is well ordered by [lemma 3.2](#), and $\mathcal{E}(G)$ is a subset of it.

Sufficiency. Assume $E = \mathcal{E}(G)$ is well ordered and put $M = \langle E \rangle_{\mathbb{N}}$. We construct a global $F \in \mathcal{G}_{r,\Gamma}(U_0)$ and local $Q_a \in \mathcal{G}_{r,\Gamma}(D_a)$ with

$$F = P_a Q_a \quad (6.3)$$

and every nonconstant coefficient supported in M . Write $P_a = I_r + \sum_{\gamma > 0} p_{a,\gamma} t^\gamma$, $F = I_r + \sum_{\gamma > 0} f_\gamma t^\gamma$, $Q_a = I_r + \sum_{\gamma > 0} q_{a,\gamma} t^\gamma$. At a positive $\gamma \in M$, with all smaller coefficients constructed, define on W_a

$$R_{a,\gamma} = p_{a,\gamma} + \sum_{\substack{\eta+\theta=\gamma \\ \eta, \theta > 0}} p_{a,\eta} q_{a,\theta}, \quad (6.4)$$

a finite sum. Apply [lemma 4.3](#) entry by entry to the prescribed ordinary negative Laurent matrices $\pi_- R_{a,\gamma}$. It supplies a single matrix $f_\gamma \in \text{Mat}_r(\mathcal{O}(U_0))$ such that $f_\gamma - R_{a,\gamma}$ extends holomorphically across D_a for every a ; set

$$q_{a,\gamma} = f_\gamma - R_{a,\gamma}. \quad (6.5)$$

The prescribed zero value at z_* can be imposed on every f_γ .

Recursion on the well-ordered set $M \setminus \{0\}$ constructs all coefficients. The scalar analytic step changes coefficient functions but introduces no new Hahn exponents, and every coefficient of every Q_a is holomorphic on its original disk. The resulting Hahn matrices satisfy [\(6.3\)](#) coefficientwise and are invertible by [lemma 3.4](#). Finally $G_a = P_a H_a$ and $F = P_a Q_a$ give $G_a = F(H_a^{-1} Q_a)^{-1}$; take $B_0 = F$ and $B_a = H_a^{-1} Q_a$. The support of $H_a - I_r$ lies in the monoid generated by the original support of $G_a - I_r$ and the support of $Q_a - I_r$ lies in M ; a finite union of two well-ordered positive sets generates a legitimate monoid, so B_a is a well-defined section on D_a . There is no requirement that the disk supports have a globally well-ordered union. $\square$

6.1 Why the converse is sharp

An infinite puncture set does not by itself prevent a solution: at each fixed exponent the ordinary analytic problem is always solvable, and the obstruction is not a divergent ordinary sum, since the ordinary coefficient construction uses normal convergence independent of the formal exponents. The failure occurs exactly when the *necessary normalized polar information* cannot fit into a single well-ordered support.

The test cannot be replaced by $\bigcup_a \text{supp}(G_a - I_r)$. First, G_a may itself be holomorphic on D_a , in which case $P_a = I_r$ and the solution $B_0 = I_r$, $B_a = G_a^{-1}$ exists even if those raw supports have a descending union. Second, and less obviously, discarding all raw holomorphic coefficients is also not enough: noncommutative products between discarded and retained terms can create essential polar information. That is the content of [section 8](#), and it is why [section 12](#) exists.

6.2 The space of solutions

Proposition 6.3 (Right torsor of splittings). *For a polar-admissible datum, all splittings are obtained from any one by*

$$(B_0, B_a) \longmapsto (B_0 T, B_a T|_{D_a}), \quad T \in \mathcal{G}_{r,\Gamma}(X),$$

and the action is free and transitive. Under the normalization $B_0(z_*) = I_r$, use the subgroup with $T(z_*) = I_r$.

Proof. Right multiplication by a common T preserves $B_0 B_a^{-1}$. Conversely two splittings give $B_0^{-1} B'_0 = B_a^{-1} B'_a$ on W_a ; the central quotient extends through every disk by that equality, and the sheaf observation of [section 2.2](#) yields a global positive section T with one legitimate support on connected X . It is the unique element relating the two splittings. $\square$

This describes all solutions once a splitting exists. It is not a classification of the nonsplit objects; see [section 21](#).

7 Invariance, and the scalar case that is recovered

Corollary 7.1 (Removal of arbitrarily supported regular data). *For independently chosen $R_a \in \mathcal{G}_{r,\Gamma}(D_a)$, the families (G_a) and $(G_a R_a)$ have the same normalized polar factors and the same solvability status. No well-ordering condition on $\bigcup_a \text{supp}(R_a - I_r)$ is needed.*

Proof. Immediate from [proposition 5.2](#) and [theorem 6.2](#). $\square$

This is stronger than invariance under finitely many transformations: the infinitely many regular parts are genuinely irrelevant, but only *after* they have been removed in the correct multiplication order.

Corollary 7.2 (Coordinate independence of admissibility). *Changing the local coordinates, or shrinking the disks, can change individual normalized polar coefficients but cannot change whether their global support union is well ordered.*

Proof. Existence of the splitting in [theorem 6.2](#) is invariant under these changes. Shrinking a disk cannot turn a previously nonextendible germ into an extendible one: a local splitting on smaller disks yields extension of the same central matrix, hence a splitting on the original disks by analytic continuation of $G_a^{-1}B_0$. Alternatively, re-expressing a previously constructed polar factor in the new coordinate and applying [theorem 5.1](#) puts every new support in the monoid generated by the old support union, and the inverse coordinate change gives the converse. $\square$

Corollary 7.3 (Absoluteness under ordered extension). *Let $\Gamma \hookrightarrow \Delta$ be an order-preserving embedding of set-sized ordered abelian groups. A datum defined over Γ splits in the positive matrix category over Δ if and only if it splits over Γ .*

Proof. The recursive local polar factor computed in Γ is also a factor in Δ and remains the unique normalized factor, so its support union is the image of $\mathcal{E}(G)$. An order embedding preserves and reflects well-ordering of a subset. Apply [theorem 6.2](#) in both groups. $\square$

In particular the obstructions below cannot be removed by allowing more surreal exponents in the same integral positive-matrix problem. This is not a claim about passing to another analytic category, and in particular not a claim about admitting negative-valuation gauges; see [section 10](#) and item 2 of [section 21](#).

Proposition 7.4 (Recovery of the rank-one singular-support test). *Let $r = 1$, put $L_a = \log G_a$, and define the logarithmic singular support*

$$\Sigma(G) = \bigcup_a \text{supp}(\pi_- L_a).$$

Then

$$\mathcal{E}(G) \text{ well ordered} \iff \Sigma(G) \text{ well ordered.} \quad (7.1)$$

Proof. By (5.5), $\log P_a = \pi_- L_a$ is supported in the positive monoid generated by $\mathcal{E}(G)$, which gives the forward direction; conversely exponentiation puts every $P_a - I_1$ inside the monoid generated by $\Sigma(G)$. Apply [lemma 3.2](#). $\square$

Combined with [theorem 6.2](#), (7.1) says that in rank one our criterion is exactly the exact Cousin criterion already proved in the repository’s global-divisor report — its Theorem 13.1, stated there in terms of the singular support of the additive data on a puncture-and-disk cover [2]. We cite that theorem as the case we recover; it is not reproved here, and no priority for the scalar statement is claimed. What is new is that the correct matrix replacement for $\pi_- \log$ is the normalized polar factor and not any entrywise logarithm: no analogous entrywise formula is substituted for Pol in $\text{rank} \geq 2$, precisely because $\log(AB) \neq \log A + \log B$ ([remark 2.2](#)), and [proposition 8.4](#) measures the error.

8 A rank-three obstruction invisible in the raw poles

This section contains the example that forces the distinction of [section 12](#). Read it before generalizing either criterion.

8.1 The closed factorization in the Heisenberg group

Proposition 8.1 (Closed Heisenberg factorization). *Let*

$$G = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad a, b, c \in \mathcal{J}_\Gamma(D^*),$$

and write $a = a_- + a_+$, $c = c_- + c_+$ by coefficientwise Laurent projection. Then

$$\text{Pol}(G) = \begin{pmatrix} 1 & a_- & \pi_-(b - a_-c_+) \\ 0 & 1 & c_- \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{Reg}(G) = \begin{pmatrix} 1 & a_+ & \pi_+(b - a_-c_+) \\ 0 & 1 & c_+ \\ 0 & 0 & 1 \end{pmatrix}. \quad (8.1)$$

Proof. All products are legitimate, being finite expressions in finitely many Hahn series. Direct multiplication verifies $\text{Pol}(G) \text{Reg}(G) = G$, the left factor is polar and the right factor is holomorphic on the full disk, so uniqueness in [theorem 5.1](#) identifies them as the canonical factors. $\square$

The subtraction of a_-c_+ is the essential term: a product of a *polar* coefficient of one entry with a *holomorphic* coefficient of another. Its exponent is a sum of two input exponents that need not appear in the input.

8.2 The obstructed datum

Take $X = \mathbb{C}$, $A = \{n : n \geq 2\}$, $D_n = \{|z - n| < 1/4\}$, $\Gamma = \mathbb{Q}$, and $u_n = z - n$. Consider

$$G_n = I_3 + \frac{t}{u_n} E_{12} + t^{1/n} E_{23}, \quad (8.2)$$

with E_{ij} the matrix units. Each G_n is a finite Hahn expression, has determinant exactly one, and reduces to I_3 . By (8.1),

$$P_n = I_3 + \frac{t}{u_n} E_{12} - \frac{t^{1+1/n}}{u_n} E_{13}, \quad H_n = I_3 + t^{1/n} E_{23}, \quad (8.3)$$

and the product $P_n H_n$ has zero $(1, 3)$ entry because its two contributions cancel.

Theorem 8.2 (Hidden rank-three obstruction). *The matrices (8.2) admit no splitting by general positive 3×3 matrices. Their abelianized upper-unitriangular cocycle does split. Their raw nonremovable coefficient support is the singleton $\{1\}$, and their determinant cocycle is identically one.*

Proof. The normalized polar support contains $\{1 + 1/n : n \geq 2\}$, a strictly decreasing infinite sequence, hence is not well ordered, and theorem 6.2 rules out a splitting even in the full positive matrix group.

The abelianization of the upper-unitriangular 3×3 group remembers the $(1, 2)$ and $(2, 3)$ entries and adds them componentwise. Let $\phi \in \mathcal{O}(U_0)$ have principal part $1/(z - n)$ at every $n \geq 2$ and no singularities outside A ; such ϕ exists by lemma 4.2, one explicit choice being

$$\phi(z) = \sum_{n=2}^{\infty} \left(\frac{1}{z - n} + \frac{1}{n} \right), \quad (8.4)$$

whose n th summand is $O_K(n^{-2})$ on each compact set away from A . The $(1, 2)$ component splits using the central function $t\phi$ and the disk functions $t(\phi - 1/u_n)$; the $(2, 3)$ component splits using central function 0 and disk functions $-t^{1/n}$. Finally the only original singular coefficient is E_{12}/u_n at exponent 1, and unitriangular matrices have determinant one. $\square$

The example is stronger than a determinant obstruction, and stronger even than the statement that a noncommutative cocycle has nontrivial abelianization: the abelianized problem *is* solvable, and the obstruction appears only when its solutions are required to fit together in the original multiplication law. The obstruction also lives in a matrix entry that is identically zero in the input.

8.3 Adding a singular term repairs the problem

Example 8.3 (The repair, which worsens the raw support). Define

$$\tilde{G}_n = I_3 + \frac{t}{u_n} E_{12} + t^{1/n} E_{23} + \frac{t^{1+1/n}}{u_n} E_{13}. \quad (8.5)$$

Then

$$\tilde{G}_n = \left(I_3 + \frac{t}{u_n} E_{12} \right) (I_3 + t^{1/n} E_{23}),$$

so $\text{Pol}(\tilde{G}_n) = I_3 + tE_{12}/u_n$ and $\mathcal{E}(\tilde{G}) = \{1\}$, and a global splitting exists. Explicitly, with ϕ as in (8.4), put $F = I_3 + t\phi E_{12}$,

$$Q_n = I_3 + t \left(\phi - \frac{1}{u_n} \right) E_{12}, \quad \tilde{H}_n = I_3 + t^{1/n} E_{23};$$

then $F = \text{Pol}(\tilde{G}_n)Q_n$ and $\tilde{G}_n = F(\tilde{H}_n^{-1}Q_n)^{-1}$, all disk factors being holomorphic on D_n . Yet the raw singular support of (8.5) *contains* the descending set $\{1 + 1/n\}$.

So (8.2) has raw singular support $\{1\}$ and is obstructed, while (8.5) has a descending raw singular support and is solvable. Raw singular support is therefore neither a sufficient nor a necessary test, in the strongest possible sense: the two directions fail on data that differ by one term.

8.4 Why an entrywise logarithm is not a repair

Proposition 8.4 (Logarithm diagnostic). *The matrix logarithm of (8.2) is finite, namely*

$$\log G_n = \frac{t}{u_n} E_{12} + t^{1/n} E_{23} - \frac{1}{2} \frac{t^{1+1/n}}{u_n} E_{13},$$

so exponentiating its negative Laurent part alone gives the $(1, 3)$ coefficient $-\frac{1}{2}t^{1+1/n}/u_n$, whereas the correct polar coefficient in (8.3) is $-t^{1+1/n}/u_n$.

Proof. $G_n - I_3$ is nilpotent of order three, so the series of lemma 3.4 terminates: with $a = t/u_n$ and $c = t^{1/n}$ one has $(G_n - I_3)^2 = ac E_{13}$ and $(G_n - I_3)^3 = 0$, giving $\log G_n = (G_n - I_3) - \frac{1}{2}ac E_{13}$, which is the display. Comparison with (8.3) gives the factor $\frac{1}{2}$. The symbolic check is recorded in the companion code (section 20). $\square$

Thus even the true matrix logarithm is not a substitute for the factorization: one still needs the noncommutative correction, and the global criterion is built from the actual normalized factor rather than from an *ad hoc* linearization. Note also that the discrepancy is only a factor, not a change of exponent, so a reader who used the logarithm would find the same descending set here — and example 8.3 shows that agreement is an accident of this example, not a method.

9 Obstructions at arbitrary nilpotent depth

9.1 A chain identity

Lemma 9.1 (Chain identity). *Let $r \geq 3$, let a be a negative Laurent Hahn function with positive support, and let $c_2, \dots, c_{r-1}$ be positive Hahn constants. Put*

$$C = \sum_{j=2}^{r-1} c_j E_{j,j+1}, \quad H = I_r + C, \quad G = I_r + C + a E_{12}.$$

Since $C^{r-1} = 0$,

$$P = GH^{-1} = I_r + a E_{12} (I_r + C)^{-1} = I_r + \sum_{k=2}^r (-1)^{k-2} a \left(\prod_{j=2}^{k-1} c_j \right) E_{1k}, \quad (9.1)$$

the empty product for $k = 2$ being 1, and (9.1) is the normalized polar factorization $P = \text{Pol}(G)$, $H = \text{Reg}(G)$.

Proof. The finite geometric inverse of $I_r + C$ gives the displayed signs. Every nonconstant coefficient of P is a negative Laurent part, because each is a constant multiple of a , while H is holomorphic on the disk. Uniqueness in theorem 5.1 identifies the factors. $\square$

9.2 The exponent design

For $r \geq 4$ and $n \geq 2$ set

$$a_n = \frac{t}{z - n}, \quad c_{2,n} = t^{1-1/n}, \quad c_{j,n} = t \ (3 \leq j \leq r-2), \quad c_{r-1,n} = t^{2/n}, \quad (9.2)$$

the middle range being empty for $r = 4$. Let C_n and G_n be the matrices of [lemma 9.1](#) built from these data. Every G_n is finite, upper unitriangular, and has exactly one row singular matrix coefficient, namely $E_{12}/(z - n)$ at exponent 1. For $3 \leq k \leq r - 1$ the exponent in $(P_n)_{1k}$ is

$$\lambda_{k,n} = k - 1 - \frac{1}{n}, \quad (9.3)$$

whereas the exponent in the final entry is

$$\lambda_{r,n} = r - 2 + \frac{1}{n}. \quad (9.4)$$

For each fixed $k < r$ the exponents (9.3) increase with n and their set has order type ω ; there are finitely many such sets, so their union with $\{1\}$ is well ordered. The final exponents (9.4) strictly decrease.

Theorem 9.2 (Arbitrarily deep central support obstructions). *For each integer $r \geq 3$ there is a positive upper-unitriangular cocycle on the puncture-and-disk cover of $\mathbb{C}$ with $A = \{2, 3, \dots\}$, over $\Gamma = \mathbb{Q}$, such that:*

- (i) *its original singular coefficient support is $\{1\}$ and its determinant is one;*
- (ii) *its image in the quotient of the upper-unitriangular group by $Z_r = \{I_r + cE_{1r} : c \in \mathcal{J}_\Gamma\}$ is a coboundary;*
- (iii) *it is not a coboundary in the full group of positive $r \times r$ matrices;*
- (iv) *it remains nonsplit after every ordered value-group extension.*

For $r = 3$ use (8.2); for $r \geq 4$ use (9.2) with [lemma 9.1](#).

Proof. Property (i) was checked directly above. [Lemma 9.1](#) and (9.4) show that the normalized polar support contains a strictly descending infinite sequence, so [theorem 6.2](#) proves (iii) and [corollary 7.3](#) proves (iv).

For (ii), let P'_n be obtained from P_n by deleting its $(1, r)$ entry. Their global support union is well ordered by (9.3), the case $r = 3$ leaving only exponent 1. Every $P'_n - I_r$ belongs to the row-one algebra

$$V = \text{span}\{E_{12}, \dots, E_{1,r-1}\}, \quad V^2 = 0,$$

so the gluing problem for the P'_n is additive and splits by the scalar construction with this one well-ordered support union: solve the ordinary principal-part problem for each coefficient of each allowed row entry, using [lemma 4.2](#). This gives F' on U_0 and Q'_n on D_n , valued in $I_r + V$, with $F' = P'_n Q'_n$. In the quotient by Z_r one has $\bar{P}_n = \bar{P}'_n$, and since $G_n = P_n H_n$ with H_n holomorphic,

$$\bar{G}_n = \bar{F}' (\bar{H}_n^{-1} \bar{Q}'_n)^{-1}$$

provides the required coboundary. All factors have individually legitimate positive supports. No matrix representative for the quotient is silently assumed: the bars denote the ordinary group quotient by the central subgroup. $\square$

Here Z_r is the central subgroup with $c \in \mathcal{J}_\Gamma$. Over $\Gamma = \mathbb{Q}$ one has $\mathcal{J}_\Gamma^m = \mathcal{J}_\Gamma$ for every positive integer m , since a nonzero section of least exponent $\delta > 0$ factors into $m - 1$ monomials $t^{\delta/m}$ and one remaining positive section. The elementary commutator identity $[I + aE_{ij}, I + bE_{jk}] = I + abE_{ik}$ then shows that Z_r is the final nonzero term of the lower central series of the positive upper-unitriangular group, and every further lower-central quotient factors through the quotient in (ii). So all those lower-depth tests pass, and the full obstruction first appears at depth $r - 1$, for arbitrarily large finite r .

Remark 9.3 (What the depth assertion means). The theorem does not say that a scalar equation cannot detect the final obstruction once a noncommutative normal form has been found — the final coefficient is of course scalar. It says that discarding the last central level *before* normalizing loses an obstruction created by the multiplication law. The new information sits in a genuinely higher noncommutative compatibility condition.

10 Integral vector bundles and local invisibility

10.1 From positive matrices to actual vector bundles

The transition matrices G_a define a locally free rank- r sheaf $\mathcal{E}_G$ over $\mathcal{I}_\Gamma$ on the ordinary surface X . Its ordinary reduction is canonically the trivial rank- r holomorphic bundle, because every transition reduces to I_r , and a positive splitting is a trivialization compatible with that reduced framing.

Proposition 10.1 (For triviality, the reduced framing costs nothing). *For a bundle defined by positive transition matrices the following are equivalent, on an arbitrary ordinary complex base and not only on a star cover:*

- (i) $\mathcal{E}_G$ is trivial as an $\mathcal{I}_\Gamma$ -module;
- (ii) its transition cocycle splits by positive matrices.

Proof. A positive splitting gives a trivialization. Conversely, suppose $A_i \in \mathrm{GL}_r(\mathcal{I}_\Gamma(U_i))$ satisfy $G_{ij} = A_i A_j^{-1}$ and reduce modulo $\mathcal{I}_\Gamma$. Since $\mathrm{red} G_{ij} = I_r$, the ordinary invertible matrices $\mathrm{red} A_i$ agree on overlaps and glue to $A \in \mathrm{GL}_r(\mathcal{O}(X))$. Right multiplication by the ordinary matrix A^{-1} produces $B_i = A_i A^{-1}$ with reduction I_r , hence $B_i \in \mathcal{G}_{r,\Gamma}(U_i)$ and $G_{ij} = B_i B_j^{-1}$. $\square$

Corollary 10.2 (Nontrivial determinant-one integral bundles). *The examples of [theorems 8.2](#) and [9.2](#) define nontrivial locally free $\mathcal{I}_\Gamma$ -modules even after forgetting the upper-triangular flag and the chosen reduced framing. Their ordinary reductions and determinant line bundles are trivial, the transition determinants being exactly one and not merely one modulo positive exponents.*

Proof. A general integral trivialization would give a positive matrix splitting by [proposition 10.1](#), contradicting [theorem 6.2](#). $\square$

10.2 Every bounded restriction is trivial

Corollary 10.3 (Exhaustion-invisible nontriviality). *Each integral bundle in [corollary 10.2](#) restricts trivially to every bounded ordinary domain in $\mathbb{C}$. In particular it is trivial on every disk of an increasing exhaustion of the plane, but not on the plane itself.*

Proof. A bounded domain meets only finitely many of the fixed disks D_n , so on the restricted cover there are only finitely many nonidentity transition matrices, each with legitimate positive support, and their union is well ordered. Every ordinary plane domain is Stein, so [theorem 11.1](#) applies. Equivalently, apply the finite-puncture construction directly and absorb transitions on pieces containing no puncture into holomorphic local gauges. $\square$

The failure is compatibility of infinitely many local trivializations, not existence on any one bounded domain. Consequently a numerical experiment involving finitely many punctures cannot establish global triviality: it will necessarily miss these examples. This is an explicit limit on what any implementation can decide, and it is recorded again as item 4 of [section 21](#).

10.3 The localization boundary

Let $\mathcal{H}_\Gamma = \mathcal{O}((t^\Gamma))$ be the sheaf in which arbitrary monomial shifts are allowed; the natural map $\mathcal{I}_\Gamma \rightarrow \mathcal{H}_\Gamma$ extends scalars from $\mathcal{E}_G$ to a rank- r $\mathcal{H}_\Gamma$ -module. We have *not* proved that the resulting module is nontrivial, and we decline the claim rather than gesture at it. The reduction argument of [proposition 10.1](#) cannot be repeated over $\mathcal{H}_\Gamma$: there is no reduction ring homomorphism $\mathcal{H}_\Gamma \rightarrow \mathcal{O}$ killing positive monomials, because those monomials are units in $\mathcal{H}_\Gamma$.

Accordingly the precise conclusions of Part I are nonexistence of a positive coherent matrix splitting and nontriviality of the corresponding *integral* coherent vector bundle. Whether negative-valuation changes of frame can trivialize particular higher-rank examples is a separate question. The scalar report proves more in rank one by a special leading-unit normalization [\[2\]](#); no matrix analogue of that argument is assumed here.

This distinction matters for the surcomplex interpretation. Integral models describe holomorphic matrices with a defined ordinary reduction and infinitesimal corrections. They are not the entire unrestricted matrix geometry over $\mathbf{SC}$, and our theorems are strong within the former category without conflating the two.

11 Positive rigidity on Stein manifolds

11.1 An arbitrary-cover theorem with a support hypothesis

Let X be a connected Stein manifold and $(U_i)_{i \in I}$ a set-indexed open cover; on disconnected intersections, support statements refer to all connected components. Suppose

$$G_{ij} \in \mathcal{G}_{r,\Gamma}(U_i \cap U_j), \quad G_{ij}G_{jk} = G_{ik}, \quad G_{ii} = I_r.$$

Call the cocycle *globally support-admissible* if

$$E = \bigcup_{i,j} \text{supp}(G_{ij} - I_r) \quad (11.1)$$

is well ordered. This is a sufficient condition *on a presentation*, not an invariant necessary condition: regular local gauges may destroy it without destroying triviality. It must not be conflated with the invariant $\mathcal{E}(G)$ of [definition 6.1](#), nor with E_ρ of [section 14](#).

Theorem 11.1 (Support-controlled Stein splitting). *A globally support-admissible positive matrix cocycle on a Stein manifold splits on its original cover: $G_{ij} = B_i B_j^{-1}$ with $B_i \in \mathcal{G}_{r,\Gamma}(U_i)$. All supports $\text{supp}(B_i - I_r)$ can be contained in the one monoid $\langle E \rangle_{\mathbb{N}} \setminus \{0\}$.*

Proof. Set $M = \langle E \rangle_{\mathbb{N}}$ and solve $G_{ij}B_j = B_i$ coefficientwise with $B_i = I_r + \sum_{\gamma \in M \setminus \{0\}} b_{i,\gamma} t^\gamma$. Assume the equations hold at all exponents below γ ; the coefficient equation at γ is

$$b_{i,\gamma} - b_{j,\gamma} = g_{ij,\gamma} + \sum_{\substack{\eta+\theta=\gamma \\ \eta,\theta>0}} g_{ij,\eta} b_{j,\theta} =: R_{ij,\gamma}, \quad (11.2)$$

a finite sum using only earlier coefficients. The cocycle identity $G_{ij}G_{jk} = G_{ik}$ together with the equations already solved below γ implies $R_{ij,\gamma} + R_{jk,\gamma} = R_{ik,\gamma}$ on triple overlaps; expand $G_{ij}(G_{jk}B_k - B_j) + (G_{ij}B_j - B_i) = G_{ik}B_k - B_i$ at exponent γ and note that all positive multipliers of a lower defect vanish by induction.

Thus $R_{ij,\gamma}$ is an ordinary additive cocycle with values in $\mathcal{O}^{r^2}$. Cartan’s theorem B gives $H^1(X, \mathcal{O}^{r^2}) = 0$ [15, 16], so it has a coboundary $b_{i,\gamma}$ on the original cover; this is again the torsor interpretation of first cohomology and assumes nothing about coherence of the Hahn sheaf. Recursion on $M \setminus \{0\}$ gives the required matrices with a common support bound, and positive inversion proves they are invertible. No new ordinary domains and no new Hahn exponents are introduced. $\square$

11.2 Deformations of a nontrivial ordinary bundle

The principle is not limited to a trivial ordinary reduction. Let $\mathcal{V}$ be an ordinary holomorphic rank- r bundle on X with transition matrices g_{ij}^0 , and suppose Hahn transitions G_{ij} have reduction g_{ij}^0 and are invertible over $\mathcal{A}_\Gamma$. The relative transitions $G_{ij}(g_{ij}^0)^{-1} - I_r$ then have positive support on overlaps.

Theorem 11.2 (Support-controlled deformation rigidity). *If the union of these relative positive supports is well ordered and X is Stein, there exist positive local matrices B_i with*

$$G_{ij} = B_i g_{ij}^0 B_j^{-1}, \quad (11.3)$$

and the nonconstant supports of all B_i lie in the monoid generated by the given support union. Thus the integral Hahn deformation is isomorphic, by an isomorphism reducing to the identity, to the scalar extension of its ordinary bundle.

Proof. Proceed as in theorem 11.1, now solving $G_{ij}B_j = B_i g_{ij}^0$. At a fixed positive exponent, multiply the residual equation on the right by $(g_{ij}^0)^{-1}$; the unknown coefficients appear as $b_{i,\gamma} - g_{ij}^0 b_{j,\gamma} (g_{ij}^0)^{-1}$. The residual is an additive cocycle for the ordinary endomorphism sheaf $\mathcal{E}nd(\mathcal{V})$, the cocycle identity following from $G_{ij}G_{jk} = G_{ik}$ and the lower solved equations exactly as before. That sheaf is locally free, hence coherent, and Cartan’s theorem B gives $H^1(X, \mathcal{E}nd(\mathcal{V})) = 0$. Solve the ordinary coefficient cocycle and continue the support-controlled recursion. All multiplications by g_{ij}^0 and its inverse occur at exponent zero and do not enlarge the exponent set. $\square$

This is a formal deformation-rigidity argument with a precise support certificate. It is not a new proof of the ordinary Oka–Grauert principle, and it is not an unconditional theorem about every Hahn-valued deformation on a noncompact base. Note that the cohomological input is acyclicity of $\mathcal{E}nd(\mathcal{V})$, not of $\mathcal{O}^{r^2}$: the theorem is about deformations of a genuinely nontrivial ordinary bundle.

11.3 A cyclic versus noncyclic dichotomy

Lemma 11.3 (Ordered-group alternative). *For a nonzero ordered abelian group Γ , the positive cone contains no infinite strictly decreasing sequence if and only if $\Gamma = \mathbb{Z}\delta$ for a least positive element δ .*

Proof. The cyclic case is immediate. Conversely, if there is no least positive element, choose a smaller positive element repeatedly. If there is a least positive δ but $\Gamma \neq \mathbb{Z}\delta$, choose a positive $\gamma \notin \mathbb{Z}\delta$; then $\gamma - n\delta > 0$ for every integer $n \geq 0$, since the first nonpositive difference would either leave an element strictly between 0 and δ or force $\gamma \in \mathbb{Z}\delta$. The sequence $\gamma, \gamma - \delta, \dots$ is strictly decreasing and positive. $\square$

Theorem 11.4 (Universal positive rigidity and its failure). *Fix a rank $r \geq 1$. Every positive rank- r cocycle on every Stein manifold is a coboundary if and only if*

$$\Gamma = \{0\} \quad \text{or} \quad \Gamma = \mathbb{Z}\delta \text{ for some } \delta > 0.$$

For every noncyclic nonzero Γ and every $r \geq 2$ there is a nonsplit determinant-one positive rank- r cocycle already on the puncture-and-disk cover of $\mathbb{C}$.

Proof. In the cyclic case every transition support is a subset of the universal well ordered positive set $\{\delta, 2\delta, \dots\}$, so its global union is automatically well ordered and [theorem 11.1](#) applies on any cover; the zero group gives only the identity cocycle.

For the converse choose a strictly decreasing positive sequence γ_n by [lemma 11.3](#) and set

$$g_n = \exp\left(\frac{t^{\gamma_n}}{z - n}\right).$$

This scalar matrix is its own normalized polar factor, every nonconstant coefficient being a multiple of a negative power of $z - n$, and its support union contains the descending set $\{\gamma_n\}$, so it is nonsplit. In rank $r \geq 2$ use

$$G_n = \text{diag}(g_n, g_n^{-1}, 1, \dots, 1). \quad (11.4)$$

Every nonconstant coefficient of this matrix is polar, so $P_n = G_n$; the same descending support prevents splitting in the full positive matrix group, while $\det G_n = 1$. For rank one the scalar example suffices. [Proposition 10.1](#) converts these examples into nontrivial integral vector bundles. $\square$

The rank-one part agrees with the Picard-group vanishing dichotomy of the repository’s global-divisor report [2], which vanishes exactly for $\Gamma = \{0\}$ or $\Gamma = \mathbb{Z}\delta$; that statement is the rank-one shadow of [theorem 11.4](#), driven by the same universal set $\{\delta, 2\delta, \dots\}$, and we cite rather than reprove it. The higher-rank conclusion here is stated for integral positive deformations, and the diagonal family (11.4) by itself supplies no nonabelian information: that is what [theorem 8.2](#) and [theorem 9.2](#) are for.

12 The two well-ordered-support criteria are not the same criterion

Both headline theorems of this report read “the union of the supports is well ordered”. The two unions are taken over different objects, and each criterion is false for the other’s object.

	Criterion P (theorem 6.2)	Criterion M (theorem 14.2)
Problem	multiplicative gluing $G_a = B_0 B_a^{-1}$ on a puncture-and-disk cover	realization of a prescribed monodromy by a global positive connection
Tested object	$\text{Pol}(G_a) - I_r$, the normalized polar factor of an <i>independent local factorization</i> of each transition matrix	$\rho(\ell_d) - I_r$, the prescribed monodromy matrices themselves, for a compatible based-meridian basis
Invariant	$\mathcal{E}(G) = \bigcup_a \text{supp}(\text{Pol}(G_a) - I_r)$	$E_\rho = \bigcup_d \text{supp}(\rho(\ell_d) - I_r)$
Raw data	raw transition support $\bigcup_a \text{supp}(G_a - I_r)$ is <i>neither necessary nor sufficient</i> : theorem 8.2 and example 8.3	the raw meridian support <i>is</i> the invariant; it is basis- and basepoint-independent by proposition 14.4
Must one factor first?	Yes. Skipping the factorization tests an object that is provably inequivalent	No. The monodromy matrices are already the right object; a local factorization of $\rho(\ell_d)$ is not part of the criterion
Invariance	under arbitrary local holomorphic right gauges (corollary 7.1), coordinate change (corollary 7.2) and ordered value-group extension (corollary 7.3)	under change of free meridian basis and base point (proposition 14.4), one fixed $\text{GL}_r(K_\Gamma)$ conjugation and ordered value-group enlargement (proposition 14.5)

The trap is concrete, and it has already been printed. In [theorem 8.2](#) the raw singular support is the single exponent $\{1\}$ and no splitting exists; in [example 8.3](#), *adding* a term whose exponents form the descending set $\{1 + 1/n\}$ among the raw poles *removes* the obstruction. So for Criterion P the raw singular support is not the invariant, and the factorization must be performed first. Nothing of the kind happens for Criterion M: there the union is over the prescribed matrices, [proposition 14.4](#) shows the condition does not depend on the presentation at all, and [example 17.1](#) obstructs precisely because the prescribed matrices themselves carry $\{1/n\}$.

Two further consequences of the difference are worth stating so that neither criterion is quoted for the other’s conclusion. First, the objects tested live in different places: $\text{Pol}(G_a)$ is a matrix of negative Laurent tails in a local coordinate at a , whereas $\rho(\ell_a)$ is a matrix of Hahn constants. Second, the two criteria are not two instances of one general theorem proved here; they are two theorems with two proofs, sharing [lemmas 3.2](#) and [3.4](#) and an ordinary Mittag–Leffler input. This report does not claim a common generalization, and [section 22](#) lists the obvious attempt as open.

Part II. Inverse monodromy: the meridian-support criterion

13 Connections, fundamental matrices, and holonomy

A positive connection on the trivial rank- r bundle over an ordinary domain U means

$$dY = \Omega Y, \quad \Omega = \sum_{\gamma \in S} t^\gamma \Omega_\gamma, \quad S \subseteq \Gamma_{>0} \text{ well ordered}, \quad (13.1)$$

where each Ω_γ is an ordinary holomorphic matrix one-form on *all* of U . In complex dimension one such connections are automatically flat: both the exterior derivative of a holomorphic one-form and the wedge of two of them vanish. As always, “positive” concerns exponents only.

Let $\tilde{U} \rightarrow U$ be the ordinary universal covering with a chosen lift $\tilde{p}$ of the base point p ; pullbacks of coefficients are understood.

Theorem 13.1 (Fundamental matrix with a support certificate). *For a connection (13.1) there is a unique positive fundamental matrix*

$$Y = I_r + \sum_{\gamma > 0} t^\gamma Y_\gamma \text{ on } \tilde{U}, \quad dY = \Omega Y, \quad Y(\tilde{p}) = I_r.$$

Every coefficient is holomorphic on all of $\tilde{U}$ and

$$\text{supp}(Y - I_r) \cup \text{supp}(Y^{-1} - I_r) \subseteq \langle S \rangle_{\mathbb{N}} \setminus \{0\}.$$

Analytic continuation determines a representation $\text{Hol}_\Omega : \pi_1(U, p) \rightarrow I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ with the same uniform support bound for every group element.

Proof. Write $T = \langle S \rangle_{\mathbb{N}} \setminus \{0\}$ and define the coefficients by well-founded recursion on T , missing coefficients of Ω being zero. At weight γ set

$$dY_\gamma = \Omega_\gamma + \sum_{\substack{\alpha + \beta = \gamma \\ \alpha, \beta > 0}} \Omega_\alpha Y_\beta, \quad Y_\gamma(\tilde{p}) = 0. \quad (13.2)$$

Only finitely many pairs contribute by [lemma 3.2](#), and $\beta < \gamma$. The right-hand side is a holomorphic matrix one-form on the simply connected Riemann surface $\tilde{U}$, so its entries have unique holomorphic primitives vanishing at $\tilde{p}$; this constructs Y_γ on the whole cover. The family has support in the well-ordered set T and satisfies the equation coefficientwise, and [lemma 3.4](#) gives its inverse with the same monoid support.

For uniqueness even against a solution with a different positive well-ordered support, take the least exponent at which two solutions differ, in the well-ordered union of their supports: at that exponent the difference has zero derivative and zero value at $\tilde{p}$, hence is zero, a contradiction.

A continued fundamental matrix differs from the original branch on a connected overlap by a constant matrix on the right, since $d(Y^{-1}\hat{Y}) = 0$ follows coefficientwise from the equation. Continuing around a based loop gives the normalized monodromy matrix, whose coefficients are values of coefficients of Y ; therefore all its positive exponents lie in T , independently of the loop. Uniqueness of continuation gives the group law with the convention of [lemma 4.4](#), and homotopy invariance follows from the universal-cover construction. $\square$

This theorem is the precise form of the *forward* map that the repository’s differential-equations report constructs from a coherent connection, recording that the exponent-zero coefficient is the ordinary monodromy matrix and that all other exponents lie in the generated monoid [\[3\]](#). Here the exponent-zero coefficient is I_r by positivity. [Theorem 14.2](#) is the converse.

13.1 The explicit iterated-integral expression

For a piecewise smooth path $c : [0, 1] \rightarrow U$ write $c^*\Omega_\gamma = A_\gamma(s) ds$. Its transport is

$$T_\Omega(c) = I_r + \sum_{n \geq 1} \sum_{\gamma_1, \dots, \gamma_n \in S} t^{\gamma_1 + \dots + \gamma_n} \int_{0 < s_1 < \dots < s_n < 1} A_{\gamma_n}(s_n) \cdots A_{\gamma_1}(s_1) ds_1 \cdots ds_n. \quad (13.3)$$

This is the usual ordered-integral solution, a classical mechanism related to iterated path integrals [\[18\]](#); the name “Dyson expansion” is used for it purely as classical mathematics, with no physical reading attached. Its meaning here is strict: the outer family is strongly summable by the finite-word clause of [lemma 3.2](#), and each individual coefficient is a *finite* sum of ordinary iterated integrals. The expression agrees with [theorem 13.1](#), since termwise differentiation gives [\(13.2\)](#); for a loop it gives Hol_Ω .

Lemma 13.2 (Triangular change of monodromy). *Suppose two positive connections Ω and Ψ agree at every exponent strictly below γ . Then their transports along any fixed path agree below γ , and for a based loop ℓ ,*

$$[t^\gamma](\text{Hol}_\Psi(\ell) - \text{Hol}_\Omega(\ell)) = \int_\ell (\Psi_\gamma - \Omega_\gamma). \quad (13.4)$$

Terms strictly above γ cannot affect coefficients at or below γ .

Proof. Compare the word expansions [\(13.3\)](#) over the well-ordered union of the two positive supports. A differing coefficient of weight at least γ cannot occur in a word of total weight less than γ ; at weight exactly γ it can occur only as the single letter of the word, every additional letter having strictly positive weight. The length-one term is the displayed ordinary period. All comparisons are finite at each exponent. $\square$

The crucial feature is that the linearized map at *every* successive weight is the same ordinary period map P of [\(4.2\)](#). Noncommutativity appears in the already determined lower-weight terms; it does not change the operator needed to solve the new correction equation.

14 The exact realization theorem

Definition 14.1 (Uniform support admissibility and σ -normalization). For $\rho : \pi_1(U, p) \rightarrow I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ with $U = \mathbb{C} \setminus D$ and $M_d = \rho(\ell_d)$, put

$$E_\rho = \bigcup_{d \in D} \text{supp}(M_d - I_r), \quad (14.1)$$

the union being over all matrix entries and all based meridians of the fixed free basis of [lemma 4.4](#). The representation is *uniformly support-admissible* when E_ρ is well ordered. For a fixed period section σ of [lemma 4.5](#), a connection is *σ -normalized* if every entry of every coefficient lies in $\text{im } \sigma \subseteq \mathcal{L}(D)$.

Theorem 14.2 (Support-sensitive Riemann–Hilbert realization: Criterion M). *Let $D \subset \mathbb{C}$ be closed and discrete, $p \in U = \mathbb{C} \setminus D$, with free based meridians fixed. For $\rho : \pi_1(U, p) \rightarrow I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ the following are equivalent:*

- (i) E_ρ is well ordered;
- (ii) ρ is the monodromy of a positive globally common-domain Hahn connection on U ;
- (iii) ρ is the monodromy of such a connection whose coefficients are meromorphic on $\mathbb{C}$ with at most simple poles in D and no other finite poles.

If these hold then for every fixed complex-linear period section σ there is exactly one σ -normalized realizer Ω^σ , and

$$\text{supp } \Omega^\sigma \subseteq \langle E_\rho \rangle_{\mathbb{N}} \setminus \{0\}. \quad (14.2)$$

Uniqueness is among all positive globally supported σ -normalized connections, not only those whose support was assumed in advance to lie in the right-hand side of (14.2).

Proof. Necessity. If $\rho = \text{Hol}_\Omega$ with $\text{supp } \Omega = S$ then [theorem 13.1](#) gives $\text{supp}(\rho(g) - I_r) \subseteq \langle S \rangle_{\mathbb{N}} \setminus \{0\}$ for every group element g , so E_ρ is a subset of one well-ordered set. This proves (ii) $\Rightarrow$ (i), and (iii) $\Rightarrow$ (ii) is immediate.

Construction. Assume (i), put $T = \langle E_\rho \rangle_{\mathbb{N}} \setminus \{0\}$, and write $M_d = I_r + \sum_{\gamma \in T} t^\gamma m_{d,\gamma}$, missing coefficients being zero. We construct $\Omega_\gamma \in \text{Mat}_r(\mathcal{L}(D))$ for $\gamma \in T$ by recursion. Suppose all lower coefficients have been constructed and form the legitimate positive connection $\Omega^{<\gamma} = \sum_{\delta \in T, \delta < \gamma} t^\delta \Omega_\delta$, whose support is a well-ordered subset of T ; the initial segment need not be finite. Define the lower-order prediction and the remaining defect by

$$h_{d,\gamma}^{<\gamma} = [t^\gamma] \text{Hol}_{\Omega^{<\gamma}}(\ell_d), \quad b_{d,\gamma} = m_{d,\gamma} - h_{d,\gamma}^{<\gamma}. \quad (14.3)$$

The prediction exists by [theorem 13.1](#); equivalently it is the finite sum of all iterated-integral words of total weight γ formed from previously constructed coefficients. Apply the period section entrywise and set

$$\Omega_\gamma = \sigma((b_{d,\gamma})_{d \in D}). \quad (14.4)$$

The datum at one exponent is an arbitrary matrix-valued element of the *product* $\mathbb{C}^D$, and [lemma 4.5](#) applies even when infinitely many of its entries are nonzero; this is why no summability across punctures is required.

The coefficient Ω_γ is meromorphic on $\mathbb{C}$ with at most simple poles in D . It corrects exactly the defect at weight γ , since $\int_{\ell_d} \Omega_\gamma = b_{d,\gamma}$ and [lemma 13.2](#) applies, and it leaves all lower weights unchanged. Transfinite recursion on the well-ordered set T therefore constructs $\Omega^\sigma = \sum_{\gamma \in T} t^\gamma \Omega_\gamma$, all of whose coefficients are holomorphic on the same domain U and whose support is a subset of T .

No limiting assertion at a limit ordinal is involved: a coefficient is simply assigned to every member of a well-ordered set.

To verify the monodromy, fix $\gamma \in T$. Coefficients at weights above γ cannot affect it by [lemma 13.2](#), and the recursion has already set it equal to $m_{d,\gamma}$. The monodromy support of the final connection lies in $\langle T \rangle_{\mathbb{N}} \setminus \{0\} = T$, so no exponent outside T appears. Thus every meridian has its prescribed monodromy, and the meridians freely generate the fundamental group by [lemma 4.4](#), proving $\text{Hol}_{\Omega^\sigma} = \rho$ and (iii).

Uniqueness. Let Ω and Ψ be two positive σ -normalized realizers. If they differ, the union of their supports is well ordered and there is a least differing exponent γ . Their holonomies are equal, so [lemma 13.2](#) gives $P(\Psi_\gamma - \Omega_\gamma) = 0$. But $\Psi_\gamma - \Omega_\gamma$ lies entrywise in $\text{im } \sigma$, and $P\sigma = \text{id}$ says that P is injective there. Hence the difference is zero, a contradiction. $\square$

Remark 14.3 (No hidden convergence over the puncture set). The construction is not a sum of one independently built differential equation per puncture. At a fixed Hahn exponent it first collapses *all* puncture data into one ordinary meromorphic one-form using the period section, and only then are these ordinary forms assembled in Hahn exponents. This move is indispensable when infinitely many punctures use the same exponent; see [example 17.2](#).

14.1 An intrinsic version of the support condition

Proposition 14.4 (Independence of the meridian basis). *For a near-identity representation ρ put $E_{\text{all}}(\rho) = \bigcup_{g \in \pi_1(U,p)} \text{supp}(\rho(g) - I_r)$. Then E_ρ is well ordered if and only if $E_{\text{all}}(\rho)$ is well ordered. Hence admissibility does not depend on the chosen free meridians or, after a base-point identification, on the base point.*

Proof. One implication follows from $E_\rho \subseteq E_{\text{all}}(\rho)$. For the other, assume E_ρ well ordered. Each group element is a finite word in the M_d and their inverses; [lemma 3.4](#) expands the inverses in the common monoid $\langle E_\rho \rangle_{\mathbb{N}}$, and multiplication preserves that monoid, so $E_{\text{all}}(\rho) \subseteq \langle E_\rho \rangle_{\mathbb{N}} \setminus \{0\}$. Changing based meridians, or identifying fundamental groups at different base points, changes the presentation of the same representation and not this condition on all its values. $\square$

This is the precise sense in which Criterion M tests an invariant of ρ itself. Contrast [section 12](#): Criterion P is *not* a condition on the raw transition data, and the analogue of [proposition 14.4](#) for $\bigcup_a \text{supp}(G_a - I_r)$ is false.

Proposition 14.5 (Fixed conjugation and enlargement cannot repair support). *For a family (M_d) over K_Γ , well-ordering of $\bigcup_d \text{supp}(M_d - I_r)$ is invariant under conjugation by any one fixed $C \in \text{GL}_r(K_\Gamma)$, and is unchanged when the family is embedded by an order-preserving value-group embedding into a larger Hahn workspace. An arbitrary conjugation need not preserve near-identity positivity; the assertion about well-ordered supports does not require that property.*

Proof. Let A and B be the supports of the finitely many entries of C and C^{-1} ; they are well ordered. If E is the union of the supports of $M_d - I_r$ then $\bigcup_d \text{supp}(C(M_d - I_r)C^{-1}) \subseteq A + E + B$, which is well ordered by the last clause of [lemma 3.2](#) when E is. The converse follows by conjugating with C^{-1} . An order embedding preserves and reflects the ordering of a subset, hence preserves and reflects its being well ordered. $\square$

[Proposition 14.5](#) for Criterion M and [corollary 7.3](#) for Criterion P are two independent proofs, for two different objects, of the same moral: enlarging the surreal workspace never repairs these obstructions.

15 Framed gauge classification and removable gauges

Definition 15.1 (Positive framed gauge). A positive gauge on U is a matrix $G \in \mathcal{G}_{r,\Gamma}(U)$. It is *framed* at p when $G(p) = I_r$ as an equality of Hahn matrices. Its inverse is a positive globally supported holomorphic matrix by [lemma 3.4](#), and it transforms a connection by

$$\Psi = dG G^{-1} + G \Omega G^{-1}. \quad (15.1)$$

Theorem 15.2 (Framed classification with controlled support). *Two positive globally Hahn-supported connections Ω and Ψ on U have the same based monodromy if and only if there is a positive framed gauge G satisfying (15.1). Such a gauge is unique, and if $E = \text{supp } \Omega \cup \text{supp } \Psi$ then*

$$\text{supp}(G - I_r) \cup \text{supp}(G^{-1} - I_r) \subseteq \langle E \rangle_{\mathbb{N}} \setminus \{0\}. \quad (15.2)$$

Consequently holonomy is a bijection between framed positive gauge classes and uniformly support-admissible near-identity representations.

Proof. Let Y_Ω, Y_Ψ be the normalized fundamental matrices on $\tilde{U}$. If the based monodromies agree, form $\tilde{G} = Y_\Psi Y_\Omega^{-1}$. Upon continuation around any loop both fundamental matrices acquire the same right factor M , so the product becomes $Y_\Psi M (Y_\Omega M)^{-1} = Y_\Psi Y_\Omega^{-1}$. Each ordinary coefficient descends to U , and the whole family already has support in $\langle E \rangle_{\mathbb{N}}$ by [theorem 13.1](#) and [lemma 3.4](#). Thus the descended matrix is a globally supported positive gauge G with $G(p) = I_r$, and differentiation gives $dG = \Psi G - G \Omega$, equivalent to (15.1).

Conversely, if a framed gauge exists then $G Y_\Omega$ solves the Ψ equation with initial value I_r , so uniqueness in [theorem 13.1](#) gives $G Y_\Omega = Y_\Psi$. Since G is single valued on U the monodromy matrices agree, and the equality also forces $G = Y_\Psi Y_\Omega^{-1}$, proving uniqueness and the support bound. The bijection now follows from [theorem 14.2](#). $\square$

The logarithmic subclass has a stronger rigidity property: although the comparison gauge is initially defined only on the punctured plane, no coefficient of it actually needs a puncture.

Theorem 15.3 (Entire extension of a logarithmic comparison gauge). *Suppose Ω and Ψ are as in [theorem 15.2](#), have the same based monodromy, and every coefficient of both lies in $\text{Mat}_r(\mathcal{L}(D))$. Then the unique framed gauge and its inverse extend coefficientwise holomorphically to all of $\mathbb{C}$; in particular $G, G^{-1} \in \mathcal{G}_{r,\Gamma}(\mathbb{C})$, with the same support bound (15.2).*

Proof. Write $G = I_r + \sum_{\gamma > 0} t^\gamma G_\gamma$. The gauge equation at weight γ reads

$$dG_\gamma = \Psi_\gamma - \Omega_\gamma + \sum_{\substack{\alpha+\beta=\gamma \\ \alpha, \beta > 0}} (\Psi_\alpha G_\beta - G_\beta \Omega_\alpha). \quad (15.3)$$

Induct on the positive support monoid in (15.2). If every lower G_β is entire, the right-hand side is an ordinary meromorphic one-form on $\mathbb{C}$ with at most simple poles in D , the sum being finite. Fix $d \in D$ and a disk containing no other point of D . The coefficient G_γ is already single valued and

holomorphic on the punctured disk, so the integral of its derivative around a small circle vanishes; hence the right-hand side of (15.3) has residue zero at d , and having at most a simple pole it is holomorphic there. Its primitive on the full disk differs from G_γ by a constant on the punctured disk, so G_γ extends holomorphically across d . Doing this for all d extends G_γ to an entire function, with no uniform bound on the sizes of coefficient functions required, and the ordinary domain after extension is the same $\mathbb{C}$ for every coefficient. Finally the geometric inverse of a positive entire family has entire coefficients and the same monoid support. $\square$

Corollary 15.4 (A global normal slice for logarithmic connections). *For a fixed σ , every positive logarithmic connection has exactly one σ -normalized representative under framed positive entire gauges, namely the connection constructed from its monodromy by (14.4).*

Proof. Its monodromy is admissible by theorem 13.1; theorems 14.2 and 15.3 give existence, and uniqueness of the normalized connection and of the framed gauge has already been proved. $\square$

Example 15.5 (A nonlogarithmic connection with trivial monodromy). On $\mathbb{C}^*$ consider the scalar connection $\Omega = -t dz/z^2$ with base point $p = 1$. Its normalized solution is $Y = \exp(t(1/z - 1))$, a positive globally supported holomorphic family on $\mathbb{C}^*$, single valued, so the monodromy is trivial. The unique framed gauge from Ω to the zero connection is $G = Y^{-1} = \exp(-t(1/z - 1))$, whose coefficient at t has a pole at 0. Therefore theorem 15.3 cannot be extended to connections with arbitrary higher-order coefficient poles.

16 Finite-puncture inversion and determinant one

Corollary 16.1 (Unique positive residue matrices for finite data). *Let $D = \{a_1, \dots, a_m\}$ be finite, fix based meridians, and let $M_j \in I_r + \text{Mat}_r(\mathfrak{m}_\Gamma)$ be arbitrary. There are unique residue matrices $R_j \in \text{Mat}_r(\mathfrak{m}_\Gamma)$ for which*

$$\Omega = \sum_{j=1}^m R_j \frac{dz}{z - a_j} \quad (16.1)$$

has based meridian monodromies M_j . If $E = \bigcup_j \text{supp}(M_j - I_r)$ then $\text{supp } R_j \subseteq \langle E \rangle_{\mathbb{N}} \setminus \{0\}$. The form has at most a simple pole at infinity, with residue $-\sum_j R_j$ there.

Proof. A finite union of well-ordered sets in a linear order is well ordered, so E is admissible. Use the explicit section (4.3) in theorem 14.2: being normalized for that section is precisely the form (16.1), and coefficientwise uniqueness gives uniqueness of all the R_j . In the coordinate $w = 1/z$ one has $dz/(z - a_j) = -dw/(w(1 - a_j w))$, proving the assertion at infinity. $\square$

This is a near-identity Hahn version of a classical Fuchsian inverse problem, not a realization result for arbitrary complex monodromy on a globally trivial bundle. The residue matrices must have positive support, a regular singularity at infinity is permitted, no assertion is made that the connection is holomorphic at infinity, and uniqueness is not claimed among arbitrary residue matrices with the same monodromy: ordinary integer shifts of exponents, which already destroy such unrestricted uniqueness, are excluded by the positive-support hypothesis.

Lemma 16.2 (Determinants and positive connections). *For the normalized fundamental matrix Y of a positive connection, $d(\det Y) = (\text{tr } \Omega) \det Y$. Consequently a traceless positive connection has determinant-one transport along every path.*

Proof. The determinant is a finite polynomial in the entries of Y , so the adjugate identity and the Leibniz rule apply coefficientwise, and Y is invertible by [theorem 13.1](#). Jacobi’s formula reduces to the displayed identity. If $\text{tr } \Omega = 0$ the determinant is constant along the normalized solution and equals 1 at the starting point. All identities involve finite coefficientwise algebra or already justified support products. $\square$

Theorem 16.3 (Traceless normalized realization). *Suppose an admissible representation in [theorem 14.2](#) takes values in $\text{SL}_r(K_\Gamma)$. For an entrywise complex-linear period section its unique normalized realizer satisfies $\text{tr } \Omega^\sigma = 0$. Conversely every traceless positive connection has monodromy in $\text{SL}_r(K_\Gamma)$, and the unique framed gauge between two traceless connections with the same monodromy has determinant one. These statements retain all the support and logarithmic-extension conclusions above.*

Proof. Use the recursion [\(14.4\)](#) and assume every lower coefficient Ω_δ traceless. Then $\Omega^{<\gamma}$ is traceless, so its monodromy N_d has determinant one by [lemma 16.2](#); by construction N_d and the target M_d agree below γ , and both have constant term I_r . In the determinant polynomial a difference at weight γ can contribute to that weight only with all other factors of weight zero, so

$$[t^\gamma](\det M_d - \det N_d) = \text{tr}(m_{d,\gamma} - h_{d,\gamma}^{<\gamma}).$$

The left side is zero, hence the defect vector in [\(14.3\)](#) is traceless at every puncture. Since the same linear section is used for each matrix entry, $\text{tr } \Omega_\gamma = \sigma((\text{tr } b_{d,\gamma})_{d \in D}) = 0$, proving the induction. The converse is [lemma 16.2](#). For a gauge satisfying $dG = \Psi G - G\Omega$, Jacobi’s formula gives $d \det G = (\text{tr } \Psi - \text{tr } \Omega) \det G = 0$, and its value at p is 1. $\square$

17 Sharp global obstructions and calibrating examples

17.1 Finite solvability does not imply global solvability

Example 17.1 (A countable special-linear obstruction). Let $D = \{1, 2, 3, \dots\}$, $\Gamma = \mathbb{Q}$, and

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_n = I_2 + t^{1/n} N.$$

Because the meridians form a free basis, this prescription defines a representation $\rho : \pi_1(\mathbb{C} \setminus D, p) \rightarrow \text{SL}_2(K_\Gamma)$. Each generator matrix has one positive exponent, is invertible, and is an entirely legitimate Hahn matrix, and every finite initial collection of the prescribed monodromies is realizable on the corresponding finitely punctured plane by [corollary 16.1](#). Nevertheless no positive globally Hahn-supported holomorphic connection on $\mathbb{C} \setminus D$ has all the prescribed monodromies.

Proof. The determinant is 1 because $N^2 = 0$ and N is strictly upper triangular. A finite set of the exponents $1/n$ is well ordered, but the full union $E_\rho = \{1, 1/2, 1/3, \dots\}$ is a strictly decreasing sequence with no least member, hence not well ordered. Nonexistence follows from [theorem 14.2](#), even without any requirement of simple poles, and [proposition 14.5](#) shows that a fixed change of matrix basis or an enlargement of the value group cannot repair it. $\square$

One-puncture realizers can even be written down:

$$\Omega_n = \frac{t^{1/n}}{2\pi i} N \frac{dz}{z - n}, \quad \exp\left(\int_{\ell_n} \Omega_n\right) = I_2 + t^{1/n} N.$$

The failure is not an inability to solve a differential equation at any one puncture; it is the impossibility of placing the entire prescription in one global support set. This is a genuine local-to-global obstruction in the chosen coefficient category, not a claim that no ordinary local system exists: the representation itself already defines an algebraic local system over K_Γ .

17.2 The union condition must not be strengthened to point-finiteness

Example 17.2 (Infinitely many punctures at one exponent). Let $D = \mathbb{Z}$, $U = \mathbb{C} \setminus \mathbb{Z}$, $\Gamma = \mathbb{Q}$, and use the same nilpotent N . Then

$$\Omega = \frac{t}{2i} N \cot(\pi z) dz \quad (17.1)$$

has monodromy $I_2 + tN$ at every positive based meridian. The family $(M_n - I_2)_{n \in \mathbb{Z}}$ is *not* strongly summable as a family indexed by punctures, because exponent 1 appears infinitely many times; it is nevertheless globally realizable.

Proof. The ordinary form $\cot(\pi z) dz$ has residue $1/\pi$ at every integer, so each meridian period of (17.1) is tN . Every product of two coefficient matrices vanishes, both being multiples of N with $N^2 = 0$, so the Dyson expansion (13.3) terminates after its length-one term and gives $I_2 + tN$. The connection has the singleton support $\{1\}$ and is holomorphic on one common domain U . Nothing in the definition asks that its periods over *different* loops form a summable family. $\square$

Together, examples 17.1 and 17.2 identify the precise strength of Criterion M. Requiring only that individual matrices be legitimate Hahn matrices is too weak; requiring a strongly summable family over punctures is too strong; a well-ordered *union* of supports is exactly right.

17.3 Admissible supports need not be locally finite in an interval

Example 17.3 (Infinitely many exponents below one bound). On $D = \{1, 2, \dots\}$ prescribe $M_n = I_2 + t^{2-1/n} N$. The support $E = \{2 - 1/n : n \geq 1\}$ is increasing of order type ω , hence well ordered, although it has infinitely many elements below 2. These data are realizable: for a fixed period section put $\alpha_n = \sigma(e_n)$ and

$$\Omega = N \sum_{n \geq 1} t^{2-1/n} \alpha_n,$$

where $e_n \in \mathbb{C}^D$ is 1 at n and 0 elsewhere. The connection has support in E , and its holonomy is exactly the prescribed family because $N^2 = 0$.

The displayed sum is a Hahn family, not an assertion that its ordinary partial sums converge in the valuation topology. In $\mathbb{Q}$ the tail valuations $2 - 1/n$ stay below 2, so those partial sums do not even approach the full family in that topology. The example also shows why our theorems supply no procedure enumerating all coefficients below a fixed bound in finitely many steps.

18 A noncommuting calculation across two valuation ranks

18.1 The targets and the first-order guess

Take $U = \mathbb{C} \setminus \{0, 1\}$ and base point $b = 1/2$, with the usual positive based meridians: a real tail to a small circle about the puncture, a positive circuit, and the reversed tail. Define

$$N_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = [N_+, N_-] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

so $N_+^2 = N_-^2 = 0$. Initially use two independent formal bookkeeping variables u, v of positive degree and prescribe

$$M_0 = I_2 + uN_+, \quad M_1 = I_2 + vN_-. \quad (18.1)$$

At the end set $u = t^\alpha$ and $v = t^\beta$ for positive, rationally independent $\alpha, \beta \in \Gamma$; the choice $\alpha = 1$, $\beta = \omega$ is the rank-two surcomplex example. Put

$$\eta_0 = \frac{dz}{2\pi iz}, \quad \eta_1 = \frac{dz}{2\pi i(z-1)}, \quad C = \frac{\log 2}{2\pi i}.$$

The obvious first-order guess is

$$\Omega^{(1)} = uN_+\eta_0 + vN_-\eta_1. \quad (18.2)$$

Its first-order periods are correct. Its mixed-order monodromy is not.

Proposition 18.1 (The first nonabelian correction). *For the connection (18.2) the coefficient of uv in the based monodromy is CH at the meridian of 0 and $-CH$ at the meridian of 1. Consequently the unique normalized realizer has expansion*

$$\Omega = uN_+\eta_0 + vN_-\eta_1 - uvCH(\eta_0 - \eta_1) + O_{\deg \geq 3}, \quad (18.3)$$

and its coefficients of u^2 and v^2 are zero. The remainder notation refers only to total degree in u, v , not to a claimed valuation bound.

Proof. Along a chosen meridian let $L_j(s) = \int_0^s \eta_j$ be the accumulated integral from the base point. The first-order transport is $I_2 + uN_+L_0 + vN_-L_1$, so the mixed term in the coefficient equation is

$$N_+N_- \int_\ell L_1\eta_0 + N_-N_+ \int_\ell L_0\eta_1. \quad (18.4)$$

For the meridian of 0, L_1 is the single-valued ordinary function $L_1(z) = (\log(1-z) - \log(1-b))/2\pi i$ on a neighbourhood containing the tail and the disk about 0 but not 1; it satisfies $L_1(b) = 0$ and $L_1(0) = C$. Integrating $d(L_0L_1)$ along the full loop gives $\int_\ell L_0\eta_1 = -\int_\ell L_1\eta_0$, because the product vanishes at both endpoints, L_0 ending at 1 but L_1 ending at 0. The residue theorem gives $\int_\ell L_1\eta_0 = L_1(0) = C$, contributions of the outgoing and incoming tails cancelling. Thus (18.4) equals $C(N_+N_- - N_-N_+) = CH$.

For the meridian of 1 use instead $L_0(z) = (\log z - \log b)/2\pi i$, which is single valued near that tail and disk with $L_0(b) = 0$ and $L_0(1) = C$; now $\int_\ell L_0\eta_1 = C$ and $\int_\ell L_1\eta_0 = -C$, so the mixed coefficient is $-CH$. The prescribed matrices (18.1) have zero uv coefficient, so the ordinary period section for $\{0, 1\}$ assigns the correcting one-form $-CH\eta_0 + CH\eta_1$ at that degree, which is (18.3). Pure second-order terms in the uncorrected transport vanish because $N_+^2 = N_-^2 = 0$, so their defects and normalized correction coefficients are zero. $\square$

In residue form the same calculation reads

$$R_0 = \frac{u}{2\pi i} N_+ - \frac{uv \log 2}{(2\pi i)^2} H + O_{\deg \geq 3}, \quad (18.5)$$

$$R_1 = \frac{v}{2\pi i} N_- + \frac{uv \log 2}{(2\pi i)^2} H + O_{\deg \geq 3}. \quad (18.6)$$

The mixed corrections cancel in the residue at infinity but are essential at the two finite poles. All displayed coefficient matrices are traceless, in agreement with [theorem 16.3](#).

18.2 The representation is genuinely nonabelian

The exact group commutator of the prescribed matrices is

$$M_0 M_1 M_0^{-1} M_1^{-1} = \begin{pmatrix} 1 + uv + u^2 v^2 & -u^2 v \\ uv^2 & 1 - uv \end{pmatrix}, \quad (18.7)$$

with determinant 1 and leading mixed term $uvH \neq 0$. So the realization theorem here is not an entrywise construction of independent scalar equations: the extra commutator in [\(18.3\)](#) is exactly the interaction that a scalar period prescription would miss. This is the Part II counterpart of the cross term $-a_- c_+$ of [proposition 8.1](#): in both parts the obstruction is manufactured by a mixed product that the abelianization cannot see.

18.3 Specialization to a genuinely higher-rank Hahn group

Set $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, $u = t$ and $v = t^\omega$. The monoid generated by 1 and ω is $\{a + b\omega : a, b \in \mathbb{N} \cup \{0\}\}$, ordered first by b and then by a , and it is well ordered. The correction in [\(18.3\)](#) occurs at exponent $1 + \omega$. No finite number of increments by 1 reaches even ω , but the well-founded recursion handles ω and $1 + \omega$ directly, with no change in the proof or in its support certificate; compare [remark 3.3](#).

The notation $O_{\deg \geq 3}$ is especially important here. A term t^3 has total degree three and yet smaller valuation than t^ω , while $t^{3\omega}$ has much larger valuation. A finite bivariate degree truncation is therefore *not* an approximation uniformly valid below a single Hahn valuation threshold. The accompanying computation checks a finite coefficient identity, not a truncation of all coefficients below $1 + \omega$.

Part III. Effectivity, verification, and boundaries

19 How to compute the invariants, and what computation cannot decide

19.1 A support-aware procedure for Criterion P

For a presented star-cover datum the mathematical procedure is:

1. at each puncture compute the normalized factor by [\(5.2\)–\(5.3\)](#), or exploit a finite matrix identity such as [\(8.1\)](#) or [\(9.1\)](#);

2. extract the union of the supports of the normalized *polar* coefficients — not the supports of the raw entries, and not just the leading valuations;
3. supply a proof of well-ordering, or a descending-sequence obstruction, for that union. In the admissible case use (6.4)–(6.5) to construct a splitting with a monoid support certificate.

These steps give an exact criterion, not a decision algorithm for an arbitrary black-box infinite family of functions. In general an input description must support a proof of the required infinite order property: checking many exponents or many punctures does not settle it, and [corollary 10.3](#) shows that a finite-puncture computation provably cannot.

19.2 Finite truncations

For finitely generated positive rational exponent monoids every bounded exponent interval is finite, so a sparse dictionary keyed by exact rational exponents suffices for a finite-cutoff implementation of the local recursion, each dictionary value being a matrix of ordinary Laurent polynomials, with π_- keeping the negative powers of u and π_+ the nonnegative ones. For a general higher-rank ordered group a bounded interval can be infinite and no such finite-cutoff assumption is justified: an implementation must either work with a certified finite dependency set or restrict its representation class. This is the computational counterpart of [remark 3.3](#), and [example 17.3](#) is a case where even the admissible answer cannot be enumerated below a fixed bound in finitely many steps.

19.3 What the Criterion M recursion actually depends on

Proposition 19.1 (Finite exponent dependence). *Fix an admissible support E , a period section σ , a target exponent $\gamma \in \langle E \rangle_{\mathbb{N}} \setminus \{0\}$, and the meridian paths. The coefficient Ω_{γ}^{σ} is determined by finitely many exponent-indexed vectors of target coefficients $(m_{d,\delta})_{d \in D}$, $\delta \in E$, and finitely many exponent patterns of ordinary products, iterated integrals, and applications of σ . The set of punctures inside one such vector may still be infinite.*

Proof. Every lower coefficient entering the defect at γ has a weight occurring as a part in a decomposition of γ into members of $\langle E \rangle_{\mathbb{N}}$, and there are finitely many such decompositions and parts by [lemma 3.5](#). Recursively replacing a constructed coefficient by its own defining defect only refines one of these finite exponent-word decompositions; equivalently all expressions are organized by finite rooted trees whose leaves are E -letters summing to γ , and there are finitely many possible leaf words by [lemma 3.2](#) and finitely many tree shapes and partitions for each fixed finite number of leaves. Applying σ may use an entire element of $\mathbb{C}^D$, so this argument says nothing about finite dependence on puncture coordinates. $\square$

A relative algorithm is therefore explicit: represent the support well enough to list the finitely many words relevant to a requested exponent, compute the lower-weight iterated-integral prediction, subtract it from the target coefficient vector, and apply the chosen ordinary period section. This does not imply that the steps are computable for every set-theoretically specified Hahn series, nor that a non-effective support can be enumerated, nor that an infinite product vector can be supplied to a finite program. For finite puncture sets the period section is explicit and the remaining tasks are ordinary symbolic or numerical integrations.

Proposition 19.2 (Compatibility with enlargement of the workspace). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be an order-preserving group embedding, with the same complex coefficient data, meridians and period section. For an admissible representation, its normalized realizer in the larger Hahn workspace is*

obtained from the original realizer by replacing t^γ with $t^{\iota(\gamma)}$, and the same compatibility holds for normalized fundamental matrices and framed comparison gauges.

Proof. Order embeddings preserve Hahn summability and addition of exponents, and the finite coefficient formulas for products, iterated integrals and ordinary primitives are unchanged, so the transfinite recursion on $\langle E \rangle_{\mathbb{N}}$ maps to the same recursion on $\langle \iota(E) \rangle_{\mathbb{N}}$ with no extra coefficients outside that monoid. Alternatively the transported family satisfies all defining equations and normalizations, so uniqueness in [theorems 13.1, 14.2 and 15.2](#) identifies it with the larger-workspace construction. $\square$

19.4 Why an infinite linear period section cannot be continuous

The choice-dependent section of [lemma 4.5](#) is not merely missing an estimate that could be supplied later. Give $\mathbb{C}^D$ its product topology and give holomorphic one-forms on U the topology of uniform convergence of the coefficient functions on compact subsets, restricted to $\mathcal{L}(D)$. The following is an auxiliary observation, not a claim of novelty for the general functional-analytic phenomenon.

Proposition 19.3 (No continuous linear period right inverse). *If D is infinite there is no continuous complex-linear right inverse $\sigma : \mathbb{C}^D \rightarrow \mathcal{L}(D)$ to P for these topologies. No such continuous linear right inverse exists even if its values are allowed to be arbitrary holomorphic one-forms on U with the specified meridian periods.*

Proof. Choose a closed ordinary disk $K \subset U$ with nonempty interior, write a one-form on U as $f(z) dz$ and put $q_K(f dz) = \sup_{z \in K} |f(z)|$. If σ were continuous and linear, the continuous seminorm $q_K \circ \sigma$ on the product $\mathbb{C}^D$ would be bounded by a constant times a maximum of finitely many coordinate absolute values, so there would be a finite $F \subset D$ and $C_0 > 0$ with

$$q_K(\sigma(v)) \leq C_0 \max_{d \in F} |v_d| \quad (v \in \mathbb{C}^D);$$

this standard product-topology bound also follows by taking a basic finite-coordinate neighbourhood on which the seminorm is less than one and scaling. Choose $d \in D \setminus F$ and take the coordinate vector e_d . The bound implies $q_K(\sigma(e_d)) = 0$, so the coefficient function of $\sigma(e_d)$ vanishes on a disk and the ordinary identity theorem on the connected domain U makes it identically zero; its period around ℓ_d is then zero, contradicting $(P\sigma(e_d))_d = 1$. $\square$

So the algebraic splitting used in the infinite-puncture normal form cannot have product-topology continuity. This does not contradict [theorem 14.2](#), whose coefficient category imposes no continuity, and it does not rule out nonlinear choices with other regularity properties, or useful linear choices on smaller weighted sequence spaces; those are different problems. For finite D the explicit section [\(4.3\)](#) is continuous and the obstruction disappears.

20 The accompanying exact and numerical checks

Two independent check suites were delivered with the two source manuscripts and are preserved unchanged in `code/` and `data/` of this report directory. Neither suite verifies a theorem; both audit finite identities, and both print their own scope disclaimer. The facts recorded in this section were confirmed by an independent run of the code, described in [appendix C](#).

Part I: exact symbolic algebra. The script `08-polar-support-matrix-cousin-verify.py` uses Python with SymPy and exact rational arithmetic, with no floating-point comparison in any checked identity. It verifies: the rank-three factors and the missing cross term, including $PH = G$, $\det = 1$, nilpotence of $G - I_3$, the repaired product of [example 8.3](#), the matrix logarithm of [proposition 8.4](#) with its $(1, 3)$ coefficient $-xy/2u$ and the exact discrepancy $xy/2u$ between the naive exponentiated-logarithm factor and the true polar factor; the chain inverse, chain factorization, row-one square-zero property and determinant for ranks 3 through 10 with symbolic c_j , together with the closed exponent formulas (9.3)–(9.4) and their monotonicity in n for ranks 4 through 12 and $n = 2, \dots, 25$; a genuinely noncommuting 2×2 series over the monoid generated by $\{1/2, 2/3\}$ truncated at cutoff 3 (sixteen exponents including zero), on which it runs the actual π_-/π_+ recursion (5.2)–(5.3) and checks $PH = G$ coefficientwise, that polar outputs are negative Laurent and regular outputs holomorphic, holomorphic right-gauge invariance $\text{Pol}(GQ) = \text{Pol}(G)$ with $\text{Reg}(GQ) = \text{Reg}(G)Q$, and a two-sided inverse; and an exact four-puncture splitting with extension of every local frame. The recorded run reports **2,899 exact checks passed** under Python 3.13.5 and SymPy 1.14.0; the independent rerun reproduced the same four PASS lines and the identical count of 2,899 under Python 3.14.4 and SymPy 1.14.0, the version banner being the only differing output line.

Part II: exact algebra plus a coefficient-ODE audit. The script `09-support-monodromy-realization-verify-examples.py` has two components. SymPy checks eight exact algebraic assertions: $N_+^2 = N_-^2 = 0$, $[N_+, N_-] = H$, both target determinants equal one, the exact group commutator (18.7) and its determinant, and tracelessness of the correction. A second component integrates the degree-graded coefficient ODE system over the bivariate degrees $(0, 0), (1, 0), (0, 1), (2, 0), (1, 1), (0, 2)$ with `scipy solve_ivp DOP853` at `rtol` $2 \cdot 10^{-12}$ and `atol` $2 \cdot 10^{-13}$, along explicit meridian paths: from $1/2$ to $1/4$, once positively around the radius- $1/4$ circle about 0, and back; and from $1/2$ to $3/4$, around the radius- $1/4$ circle about 1 from angle π to 3π , and back. It does *not* substitute floating-point surrogates for u or v . The uncorrected mixed term is compared with $\pm CH$ and the corrected transport with the target through degree two. Recorded maximum entrywise errors are $1.17 \cdot 10^{-13}$ and $1.47 \cdot 10^{-13}$ respectively at both meridians; the assertions use a looser absolute tolerance of $2 \cdot 10^{-9}$ so as not to depend on platform-specific last digits. The recorded environment is Python 3.13.5 on Linux with NumPy 2.3.5, SciPy 1.17.0 and SymPy 1.14.0.

A warning about rerunning the Part II script. That script *writes its own evidence*: it emits `verification_results.json` beside itself, via `Path(__file__).with_name`, so rerunning it in place overwrites the delivered record. Any byte-identity check performed after an in-place rerun compares two equally modified copies and proves nothing. The independent rerun reported in [appendix C](#) was therefore performed on a copy outside this report directory, and the delivered `09-support-monodromy-realization-verification-results.json` in `data/` is the original record.

What the checks do not establish. The Part I checks audit algebraic signs, multiplication order and finite coefficient identities; they do not prove the support lemma, Cartan’s theorem B, any infinite descending obstruction, an analytic existence theorem, or novelty. The Part II numerical results are observed floating-point errors, not rigorous enclosures, and test only a finite bivariate degree-two identity; the exact checks are algebraic identities in two formal variables. Neither suite tests an arbitrary Hahn field, justifies a transfinite recursion, proves any support set well ordered, verifies anything in a proof assistant, or establishes that the results are unpublished. No implication

in this report depends on the outcome of either script: the general proofs are the arguments in the body.

21 Exactly what is not claimed

The following list merges the honest limitations stated by both source manuscripts. Every one of them survives this merge, and none is weakened. Items 1 through 11 come from Part I’s scope, items 12 through 21 from Part II’s, and the boundary note at the end records the one place where the two parts make different standing assumptions.

1. Nothing here is refereed or proof-assistant verified. Both source manuscripts are self-described AI-assisted research drafts and say so on their title pages and in their READMEs.
2. Nontriviality in Part I is proved *only* over the integral sheaf $\mathcal{I}_\Gamma$. We explicitly do not claim that the examples survive after all monomials are inverted, that is over $\mathcal{H}_\Gamma$, because positive monomials become units there, no reduction homomorphism $\mathcal{H}_\Gamma \rightarrow \mathcal{O}$ exists, and the framing-normalization argument of [proposition 10.1](#) cannot be repeated. This is listed as an open question, not as a believed consequence. In particular neither criterion of this report is restated over a coefficient field in which negative-valuation changes of frame are admitted.
3. The exact criterion of [theorem 6.2](#) is only for the puncture-and-disk (star) cover, where each overlap has a distinguished isolated Laurent singularity. For general covers only the *sufficient* globally support-admissible theorem [theorem 11.1](#) is available, and its hypothesis ([11.1](#)) is a condition on a *presentation*, not an invariant necessary condition: regular local gauges can destroy it without destroying triviality.
4. Neither criterion is a decision algorithm for a black-box infinite family. An input description must support a proof of the infinite order property; checking many exponents or many punctures does not settle it, and [corollary 10.3](#) shows that finite-puncture computation provably cannot detect the Part I examples.
5. Finite-cutoff implementation is justified only for finitely generated positive rational exponent monoids. For a general higher-rank ordered group a bounded exponent interval can be infinite.
6. No classification of isomorphism classes is offered; only triviality is decided. [Proposition 6.3](#) describes all solutions once a splitting exists, not the nonsplit objects.
7. The sharp star-cover invariant does not generalize to positive-dimensional analytic singular loci: there is no one-variable Laurent projection there, and an ideal-theoretic or cohomological replacement may have nontrivial relations.
8. Credited as established mathematics and not new: the Neumann support calculus; the ordinary Laurent, Mittag–Leffler, Cartan B and Behnke–Stein inputs; and algebraic noncommutative Birkhoff factorization in complete filtered algebras. The local factorization is reproved only for explicit arbitrary-ordered-group support control. The citation of [\[12\]](#) is to the *algebra* of recursive Birkhoff factorization and to the necessity of commutator corrections; nothing in this report imports, endorses or reinterprets any application of that algebra, and no assertion here is physical, empirical or applied in any way. Likewise “positive” never means Hermitian positivity, and the normalized polar factorization is *not* the Hilbert-space polar decomposition.
9. The literature checks were targeted and are explicitly not certifications of priority. Broad searches that returned irrelevant material were not treated as evidence of absence, a repository code-search response marked incomplete was not used as evidence of absence, and no named published conjecture is claimed solved.
10. The companion code checks exact finite algebra and one finite numerical example only; see the final paragraph of [section 20](#).

11. Formal verification is future work. The local support recursion, the chain identity and the necessity directions appear well suited to formalization with Hahn series and finite matrix algebra, but the global sufficiency proofs additionally need a formal ordinary Mittag–Leffler theorem or a suitable sheaf-cohomology interface. Neither script is a substitute for such a formalization.
12. Positivity of exponents is essential throughout Part II. There is no treatment of nontrivial exponent-zero monodromy, of arbitrary ordinary residues with resonances, or of arbitrary algebraic groups.
13. [Theorem 14.2](#) is not an unrestricted classical Riemann–Hilbert theorem and not a Riemann–Hilbert correspondence on a general surcomplex domain; likewise [theorem 6.2](#) is not the unrestricted classical Cousin theorem over $\mathcal{H}_\Gamma$. The base, the universal cover and the path category are *ordinary* complex by design, not provisional notation for a fine-topological construction.
14. There is no fine-topological theory of surcomplex loops here. Convergence of Hahn partial sums in the fine topology is never asserted, and [example 17.3](#) shows that partial sums can fail to converge even in the valuation topology. The surcomplex reading is an interpretation of a Hahn model via $t^\gamma = \omega^{-\gamma}$ and does not assert an ambient surcomplex manifold.
15. The infinite-puncture period section is obtained by a vector-space splitting using the axiom of choice. It is not continuous — [proposition 19.3](#) proves that impossible — not normed, not effective and not computable; only an ordinary complex-linear splitting is claimed.
16. For infinitely many punctures, no regularity or growth condition at infinity is imposed on realizers. This is deliberate, and the freedom to use arbitrary ordinary entire correction terms at each exponent is important to the existence proof.
17. Part II does not assume that the global-support assignment is a sheaf for arbitrary covers, uses no Cartan theorem and no sheaf cohomology for a Hahn coefficient sheaf, and relies on no companion claim about Hahn Noetherianity or a Hahn Nullstellensatz.
18. Deformations of a *nontrivial ordinary connection* are open. The linearized operator there becomes a twisted period map with values in an adjoint local system rather than the scalar period map P , and the triangular proof of [lemma 13.2](#) supplies no right inverse for it. This is not to be confused with [theorem 11.2](#), which deforms a nontrivial ordinary *vector bundle* on a Stein base and does have its cohomological input available.
19. Higher-dimensional bases are open for Part II: flatness is automatic only for holomorphic one-forms on a curve, a proposed correction must satisfy the nonlinear flatness equation as well as the period conditions, and the result cannot be inferred by changing z to a tuple.
20. The Part II numerical checks are observed floating-point errors, not rigorous enclosures, and test only a finite bivariate degree-two identity.
21. The Hahn-meromorphic category of Müller–Strohmaier [21] is explicitly *not* identified with ours: there the parameter is tied to an expansion near zero on a logarithmic cover, whereas here t is an independent Hahn parameter and z the ordinary analytic variable.
22. No theorem of either part is claimed to be an instance of a common generalization proved here. The two criteria are two theorems with two proofs ([section 12](#)); they share only [lemmas 3.2](#) and [3.4](#) and an ordinary Mittag–Leffler input.

Boundary note on the sheaf convention. As recorded in [section 2.2](#), Part I proves and uses the sheaf property of the integral common-domain assignment on the ordinary base, while Part II deliberately assumes nothing of the kind. Both are correct as stated, the difference is one of standing hypotheses rather than of fact, and no theorem of Part II is strengthened here by importing Part I’s

convention.

22 Questions exposed by the results

Monomial localization. Does $\mathcal{E}_G \otimes_{\mathcal{H}} \mathcal{H}_T$ remain nontrivial for the explicit hidden and deep examples? A proof would need control of general Hahn matrix gauges, not just positive ones, and the ordinary reduction normalization is unavailable after localization. This question is intentionally not answered.

Arbitrary-cover normal forms. The polar support is exact on the puncture-and-disk cover because every overlap has a distinguished isolated Laurent singularity. An equally explicit invariant for general covers would require a substitute for that canonical local negative factor; [theorem 11.1](#) does not provide one.

Classification beyond triviality. Deciding whether a positive star-cover cocycle represents the trivial integral bundle is not classifying all isomorphism classes, which would require understanding the action of global and local gauges on *pairs* of polar families.

A common generalization of the two criteria. Both criteria say that a union of supports must be well ordered, over different objects ([section 12](#)). Is there a single statement, in some category of positive Hahn objects on a punctured base with a distinguished local normalization, of which both are instances? We do not know, and nothing in this report should be read as evidence that the naive common statement — taking the union over raw local data — is true: [theorem 8.2](#) and [example 8.3](#) refute it for Criterion P.

Deformations of a nontrivial ordinary connection. Fix an ordinary connection Ω_0 and a representation reducing to its monodromy rather than to the identity. The first correction operator is then a twisted period map into the adjoint local system. A precise extension should specify when that map has a right inverse within a chosen pole class, and should track local resonances.

Controlled growth and effective infinite data. For a prescribed weight or growth condition on residues at D , identify spaces on which a useful continuous or computable period section exists. The exact question is to characterize the residue sequence spaces admitted by the chosen coefficient growth class, and then to prove that nonlinear monodromy corrections stay inside them; arbitrary algebraic splittings need not preserve any such structure.

Higher-dimensional bases. For Stein manifolds of dimension greater than one the support-controlled rigidity theorems still apply, but the sharp star-cover invariant does not generalize, and for connections the nonlinear flatness equation enters. Both directions are open.

Formal verification. See item 11 of [section 21](#).

23 Conclusion

In the positive integral Hahn category over a set-sized ordered value group, both the multiplicative gluing problem and the inverse monodromy problem have exact answers, and in both the answer is a well-ordering condition on a union of Hahn supports. They are not the same condition.

For gluing, the correct obstruction is not the support of the input entries, nor the support of their raw principal parts, nor a determinant. One must first remove the holomorphic factors in the correct noncommutative order; the resulting normalized polar support is then an exact necessary and sufficient criterion on a puncture-and-disk cover. The explicit examples show why that normalization is consequential: a family with one singular input exponent can force a descending support at a hidden matrix entry; adding a cross term can repair the family while worsening its raw singular support; and the same phenomenon can be postponed to the final central layer of upper-unitriangular groups of arbitrarily large finite rank, yielding nontrivial determinant-one integral Hahn bundles with trivial ordinary reduction that are trivial on every bounded plane domain. At the opposite boundary, one globally well-ordered support certificate restores Stein rigidity in every finite rank; cyclic value groups supply such a certificate automatically and noncyclic ones do not.

For monodromy, the correct obstruction *is* the raw support of the prescribed matrices, and it is an invariant of the representation. Realizers can be taken coefficientwise logarithmic, a fixed linear period section selects a unique normalized one, framed positive gauges classify connections with equal monodromy, and within the logarithmic class the comparison gauge extends across every puncture. The order of the value group may have several ranks, the singularity set may be infinite, and the monodromy may be noncommutative; none of this requires a false topological limiting argument.

What both halves separate cleanly is ordinary analytic solvability, noncommutative compatibility, and generalized-series admissibility. The analytic work is done coefficientwise on the ordinary base; the Hahn support calculus is exactly what makes those ordinary constructions coexist in one surcomplex workspace; and the noncommutative correction — the cross term $-a_-c_+$ in Part I, the mixed commutator $-uv\,CH(\eta_0 - \eta_1)$ in Part II — is what the abelianization cannot see.

A A proof and support audit

The table records every place where an infinite formal expression occurs, together with the set that controls it and the ordinary analytic fact required. A support in the middle column is a *containing* support; cancellation may make the actual support smaller.

Part I constructions.

Construction	Support control	Ordinary analytic requirement
Positive matrix inverse, exp, log	Positive monoid generated by the input support; finitely many exponent words and finitely many matrix paths at each coefficient (lemmas 3.2 and 3.4)	Coefficients remain on the original domain.
Local polar factorization	$\langle \text{supp}(G - I_r) \rangle_{\mathbb{N}}$; well-founded triangular recursion	Laurent splitting on one fixed punctured disk, with no shrinking (lemma 4.1).
Necessity of Criterion P	All $P_a - I_r$ lie in the one monoid generated by $\text{supp}(B_0 - I_r)$	Right-gauge invariance and uniqueness of the local factorization.
Sufficiency of Criterion P	$\langle \mathcal{E}(G) \rangle_{\mathbb{N}}$, for the central matrix and for the Q_a	One ordinary isolated-singularity interpolation per matrix coefficient and exponent (lemmas 4.2 and 4.3).
Disk trivializers	Generated monoid of the global polar support together with that disk's own original support	Products of matrices holomorphic on that disk.
Right torsor of splittings	Finite union of the two solution supports, then positive monoid closure	Identity-theorem extension across each puncture.
Stein splitting	One monoid generated by all transition supports	Ordinary $H^1(X, \mathcal{O}^{r^2}) = 0$.
Deformation rigidity	The same positive monoid; ordinary transitions are exponent-zero coefficients	Ordinary $H^1(X, \mathcal{E}nd(\mathcal{V})) = 0$.
Rank-three obstruction	Forced descending set $\{1 + 1/n\}$	Only simple poles are used.
Depth- $(r - 1)$ obstruction	Lower partial-sum supports well ordered; final support $\{r - 2 + 1/n\}$ descends	Only simple poles and holomorphic constants are used.
Integral nontriviality	A hypothetical integral frame normalizes to a positive frame	Reduction of invertible matrices and gluing of the reduced frame.

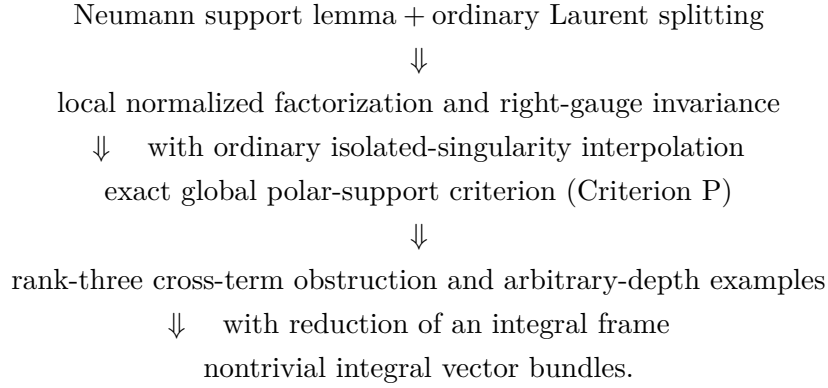
Part II constructions.

Construction	Support control	Ordinary analytic requirement
Fundamental matrix on the cover	$\langle \text{supp } \Omega \rangle_{\mathbb{N}}$, for Y and for Y^{-1} ; the same bound for <i>every</i> holonomy matrix	Holomorphic primitives on a simply connected Riemann surface.
Dyson transport	Strong summability by the finite-word clause; each coefficient a finite sum of iterated integrals	Ordinary iterated path integrals along an ordinary path.
Necessity of Criterion M	$E_{\rho} \subseteq \langle \text{supp } \Omega \rangle_{\mathbb{N}} \setminus \{0\}$, uniformly in the group element	None beyond the fundamental-matrix theorem.
Sufficiency of Criterion M	$\langle E_{\rho} \rangle_{\mathbb{N}}$; transfinite recursion with no limit-ordinal passage	Ordinary Mittag–Leffler through the period map P , plus one fixed linear section σ (axiom of choice for infinite D).
Uniqueness of the normalized realizer	Least differing exponent in a well-ordered union	Injectivity of P on $\text{im } \sigma$.
Framed gauge and its inverse	$\langle \text{supp } \Omega \cup \text{supp } \Psi \rangle_{\mathbb{N}}$	Single-valuedness on U ; descent of coefficients from the cover.
Entire extension of a logarithmic gauge	The same monoid, unchanged by the extension	A simple pole with zero residue is removable.
Positive Frobenius form	$\langle E \rangle_{\mathbb{N}}$ for H and H^{-1}	Primitives on a disk; $H_{\beta}(0) = 0$ makes H_{β}/z holomorphic.
Traceless refinement	No change of support	Finite polynomial determinant identity; one entrywise linear section.
Finite exponent dependence	Finitely many decompositions and parts inside $\langle E \rangle_{\mathbb{N}}$ (lemma 3.5)	Finitely many rooted trees and ordinary operations per exponent.

The indispensable hypotheses are these. The value group is a set and is totally ordered. Supports are well ordered, not merely bounded below. Matrix rank is finite. The puncture set is ordinary closed and discrete. Coefficient functions share their stated ordinary domains. In Part I the gluing matrices reduce to the identity, or to a specified ordinary bundle in [theorem 11.2](#); in Part II the representation is near-identity with positive support. None of these hypotheses is replaced anywhere by a fine-topological convergence assumption.

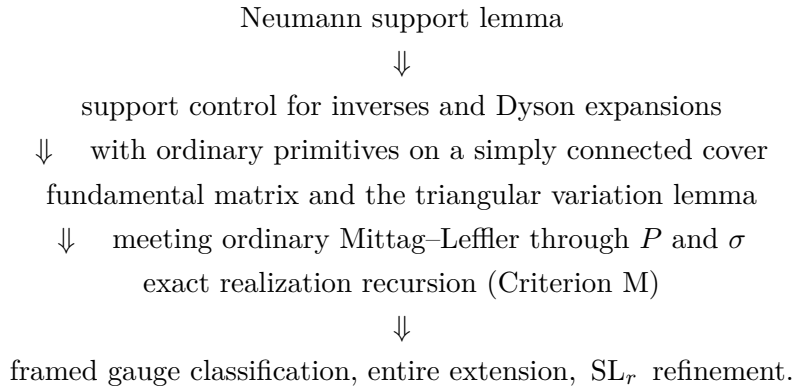
B A dependency map

The Part I chain is



The arbitrary-cover Stein theorem is a separate positive branch whose only cohomological input is ordinary coherent-sheaf acyclicity. It is not used to justify the infinite star-cover criterion on the plane, whose interpolation step has its own direct proof; it is used for the statement about all bounded restrictions and for the universal cyclic-group result.

The Part II chain is



The gauge classification uses only the fundamental matrices; its entire extension uses only single-valuedness and removability of a simple pole with zero residue. The traceless refinement adds a finite polynomial determinant identity. The counterexamples use the necessity half of [theorem 14.2](#) and explicit support orderings. No implication in either chain depends on the outcome of a verification script.

The two chains meet only at [lemmas 3.2](#) and [3.4](#) and at the shared observation that the ordinary analytic input is applied one exponent at a time. Neither criterion is derived from the other.

C Provenance, reproduction, and the merge record

C.1 Sources of this report

This report merges two independently written manuscripts, both dated September 21, 2026, both delivered with their own code and recorded output, the latter preserved unchanged in `code/` and `data/` of this directory:

- *Nonabelian Support Obstructions in Surcomplex Analysis: A Sharp Matrix Cousin Criterion and Arbitrarily Deep Unipotent Examples*, from the archive `surcomplex_nonabelian_support`; source file `08-polar-support-matrix-cousin.tex`, delivered with its README. It contributes Part I entire, the shared coefficient category and support calculus, the Laurent and isolated-singular-part inputs, the normalized polar factorization, the exact Criterion P with its invariance corollaries and torsor statement, the rank-one recovery, the rank-three obstruction with its repair and logarithm diagnostic, the chain identity and depth hierarchy, the integral vector-bundle statements with the localization boundary, and the Stein splitting, deformation rigidity and cyclic dichotomy.
- *Global Hahn Support and the Surcomplex Riemann–Hilbert Problem: Noncommutative monodromy, countably many singularities, and an exact realization obstruction*, from the archive `surcomplex_riemann_hilbert`; source file `09-support-monodromy-realization.tex`, delivered with its README and its `BUILD_AND_PROVENANCE.json`. It contributes Part II entire: strong summability and the finite-decomposition lemma, the free-meridian lemma, the period map with its linear section, the fundamental-matrix and triangular lemmas, Criterion M with its intrinsic and conjugation-invariance forms, the framed gauge classification and entire extension, the positive Frobenius form, the finite Fuchsian specialization and the SL_r refinement, the three calibrating examples, the mixed commutator correction with its $\mathbb{Q} + \mathbb{Q}\omega$ specialization, the finite-dependence and base-change propositions, and the impossibility of a continuous linear period right inverse.

Where the two manuscripts proved the same fact — the Neumann support calculus as an input, and the positive inversion, exponential and logarithm lemma — the statement is printed once, in [lemmas 3.2](#) and [3.4](#), and both members verified it independently in the same form. Where they supplied genuinely different routes to a needed ingredient, both routes are kept and marked: the ordinary Mittag–Leffler input appears twice, as the unbounded-order isolated-singular-part construction [lemmas 4.2](#) and [4.3](#) and as the linear period section [lemma 4.5](#), for the reasons given at the end of [section 4](#); and the impossibility of repair by enlarging the value group is proved twice, for two different objects, in [corollary 7.3](#) and [proposition 14.5](#). The local factorizations of [theorem 5.1](#) and [theorem 5.4](#) are likewise two different normalizations of two different objects, and neither is derived from the other here.

C.2 Repository provenance

Repository: <https://github.com/VladimirReshetnikov/Surreal>, reviewed at commit `39f2be6667ade51bca2b45daa47e289d69c09764`, inspected September 21, 2026. No repository files were changed by either source manuscript and nothing was pushed by them. The report catalogue [\[1\]](#) supplies the report map and the distinctions between coefficient rings, which both members call essential. The global-divisor report [\[2\]](#) is the source of the motivating open direction and of the scalar results we recover rather than reprove; the differential-equations report [\[3\]](#) supplies the forward monodromy map and the ordinary-paths wording. The analysis, analytic-geometry and contour packages [\[4, 5, 6\]](#) are credited by Part II as antecedents for scalar global support obstructions and for the coefficientwise differential and integration viewpoint; they are not contributions claimed here, and no draft ring theorem or Hahn sheaf-cohomology theorem from them is used in any proof.

Neither source manuscript conducted an exhaustive audit of every source manuscript in the repository, and one repository code-search response was marked incomplete; its empty result is not evidence of absence. The comparison above is with the documented contents actually reviewed.

C.3 Building this report

The source is standalone, with an internal bibliography, no external bibliography file, no graphics and no separately distributed fonts. A standard TeX Live installation builds it with

```
latexmk -pdf -interaction=nonstopmode article.tex
```

C.4 Rerunning the delivered checks

The two scripts are preserved under `code/` with their original contents and their recorded outputs under `data/`. The Part I script needs Python 3.10 or later with SymPy; the Part II script needs NumPy, SciPy and SymPy. From a working copy:

```
python code/08-polar-support-matrix-cousin-verify.py
python code/09-support-monodromy-realization-verify-examples.py
```

Two operational cautions, both verified directly rather than taken from a README.

First, the Part II script writes `verification_results.json` beside itself and therefore overwrites its own delivered record when run in place. Run it on a copy outside this directory. A byte-identity comparison performed after an in-place run compares two equally modified copies and proves nothing.

Second, the Part II README as delivered lists a file `SHA256SUMS.txt` among the archive contents. That file is *not* present in the delivered directory and is not present here, so no checksum verification of the original archive is possible from what was shipped.

An independent rerun of both suites, on copies and not in the extraction tree, found: the Part I suite exits zero and prints its four PASS lines and **SUCCESS: 2899 exact checks passed**, identical in count to the recorded run, under Python 3.14.4 with SymPy 1.14.0 against the recorded Python 3.13.5 with SymPy 1.14.0, the version banner being the only differing line; and the Part II suite exits zero with all eight exact checks and all four numerical error checks passing. The recorded Part II build metadata claims zero warnings, zero overfull or underfull boxes, zero undefined references, a 25-page document, and a Poppler render review of all pages, and records `formal_verification: false`, `priority_certified: false`, `peer_reviewed: false`. Those three flags remain false for this merged report.

References

- [1] V. Reshetnikov, *Surreal repository: report catalogue*, docs/README.md, commit 39f2be6667ade51bca2b45daa47e289d69c09764, inspected September 21, 2026.
<https://github.com/VladimirReshetnikov/Surreal/blob/39f2be6667ade51bca2b45daa47e289d69c09764/docs/README.md>.
- [2] *Global Support Obstructions: Moving Divisors, Mittag-Leffler, and a Picard Dichotomy*, AI-assisted merged research draft in V. Reshetnikov’s *Surreal* repository, docs/surcomplex/global-divisors/. Pinned commit as above; see especially the exact Cousin criterion (Theorem 13.1), the Picard-group vanishing dichotomy, and the subsection “Further precise directions”.
<https://github.com/VladimirReshetnikov/Surreal/blob/39f2be6667ade51bca2b45daa47e289d69c09764/docs/surcomplex/global-divisors/article.tex>.
- [3] *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers*, AI-assisted merged research draft in V. Reshetnikov’s *Surreal* repository,

- docs/surcomplex/differential-equations/. Pinned commit as above; see especially the theorem “Monodromy valued in $GL_d(K_\Gamma)$ ” and the ordinary-path remark following it.
<https://github.com/VladimirReshetnikov/Surreal/blob/39f2be6667ade51bca2b45daa47e289d69c09764/docs/surcomplex/differential-equations/article.tex>.
- [4] *Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$* , docs/surcomplex/analysis/README.md, same repository and commit. AI-assisted, unrefereed research documentation.
 - [5] *Infinitesimal analytic geometry over the surcomplex numbers: radius-free germs, a Nullstellensatz, and finite maps*, docs/surcomplex/analytic-geometry/README.md, same repository and commit. AI-assisted, unrefereed research documentation.
 - [6] *Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane*, docs/surcomplex/contours-and-stokes/README.md, same repository and commit. AI-assisted, unrefereed research documentation.
 - [7] B. H. Neumann, On ordered division rings, *Transactions of the American Mathematical Society* **66** (1949), 202–252. <https://www.jstor.org/stable/1990552>.
 - [8] B. Poonen, *Maximally complete fields*, L’Enseignement Mathématique (2) **39** (1993), 87–106. DOI: <https://doi.org/10.5169/seals-60414>.
 - [9] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>.
 - [10] V. Mantova and M. Matusinski, *Surreal numbers with derivation, Hardy fields and transseries: a survey*, arXiv:1608.03413. <https://arxiv.org/abs/1608.03413>.
 - [11] O. Bournez and Q. Guilmant, *Surreal fields stable under exponential and logarithmic functions*, preprint (2022), arXiv:2201.08199. Cited for normal-form and set-sized-surreal-field background only. <https://arxiv.org/abs/2201.08199>.
 - [12] K. Ebrahimi-Fard, L. Guo, and D. Kreimer, Spitzer’s identity and the algebraic Birkhoff decomposition in pQFT, *Journal of Physics A: Mathematical and General* **37** (2004), 11037–11052. Cited solely as the closest algebraic antecedent for recursive Birkhoff factorization in complete filtered noncommutative algebras and for the necessity of commutator corrections; no application of that algebra is imported here. <https://arxiv.org/abs/hep-th/0407082>.
 - [13] K. Ebrahimi-Fard, D. Manchon, and F. Patras, *A noncommutative Bohnenblust–Spitzer identity for Rota–Baxter algebras solves Bogoliubov’s recursion*, arXiv:0705.1265. <https://arxiv.org/abs/0705.1265>.
 - [14] H. Behnke and K. Stein, Entwicklung analytischer Funktionen auf Riemannschen Flächen, *Mathematische Annalen* **120** (1947/1949), 430–461. <https://doi.org/10.1007/BF01447838>.
 - [15] H. Cartan, Variétés analytiques complexes et cohomologie, in *Colloque sur les fonctions de plusieurs variables*, Brussels, 1953, 41–55. Classical source for coherent-sheaf acyclicity on Stein manifolds.
 - [16] M. M. Schmitt, *Kohomologie mit Schranken und Fortsetzung holomorpher Funktionen durch lineare stetige Operatoren*, doctoral thesis, arXiv:math/0202263 (2002). Includes a proof of Cartan’s theorem B for coherent subsheaves of finite free holomorphic sheaves and coefficientwise coboundary constructions. <https://arxiv.org/abs/math/0202263>.
 - [17] O. Wone, *The Birkhoff–Grothendieck theorem*, arXiv:2304.08037, version 2. Historical and mathematical exposition of bundle splitting and Birkhoff factorization; its usual single-loop setting differs from the infinitely many independently supported gluing data treated here. <https://arxiv.org/abs/2304.08037>.
 - [18] K.-T. Chen, *Iterated path integrals*, Bulletin of the American Mathematical Society **83** (1977), 831–879. DOI: <https://doi.org/10.1090/S0002-9904-1977-14320-6>.

- [19] P. Deligne, *Équations différentielles à points singuliers réguliers*, Lecture Notes in Mathematics, vol. 163, Springer, 1970. A classical reference point, not a theorem reproved here in its original generality. DOI: <https://doi.org/10.1007/BFb0061194>.
- [20] Y. Ilyashenko and S. Yakovenko, *Lectures on Analytic Differential Equations*, Graduate Studies in Mathematics, vol. 86, American Mathematical Society, 2008. DOI: <https://doi.org/10.1090/gsm/086>.
- [21] J. Müller and A. Strohmaier, *The theory of Hahn-meromorphic functions, a holomorphic Fredholm theorem, and its applications*, Analysis & PDE **7** (2014), no. 3, 745–770. Their analytic category is explicitly not identified with ours; see item 21 of section 21. DOI: <https://doi.org/10.2140/apde.2014.7.745>.